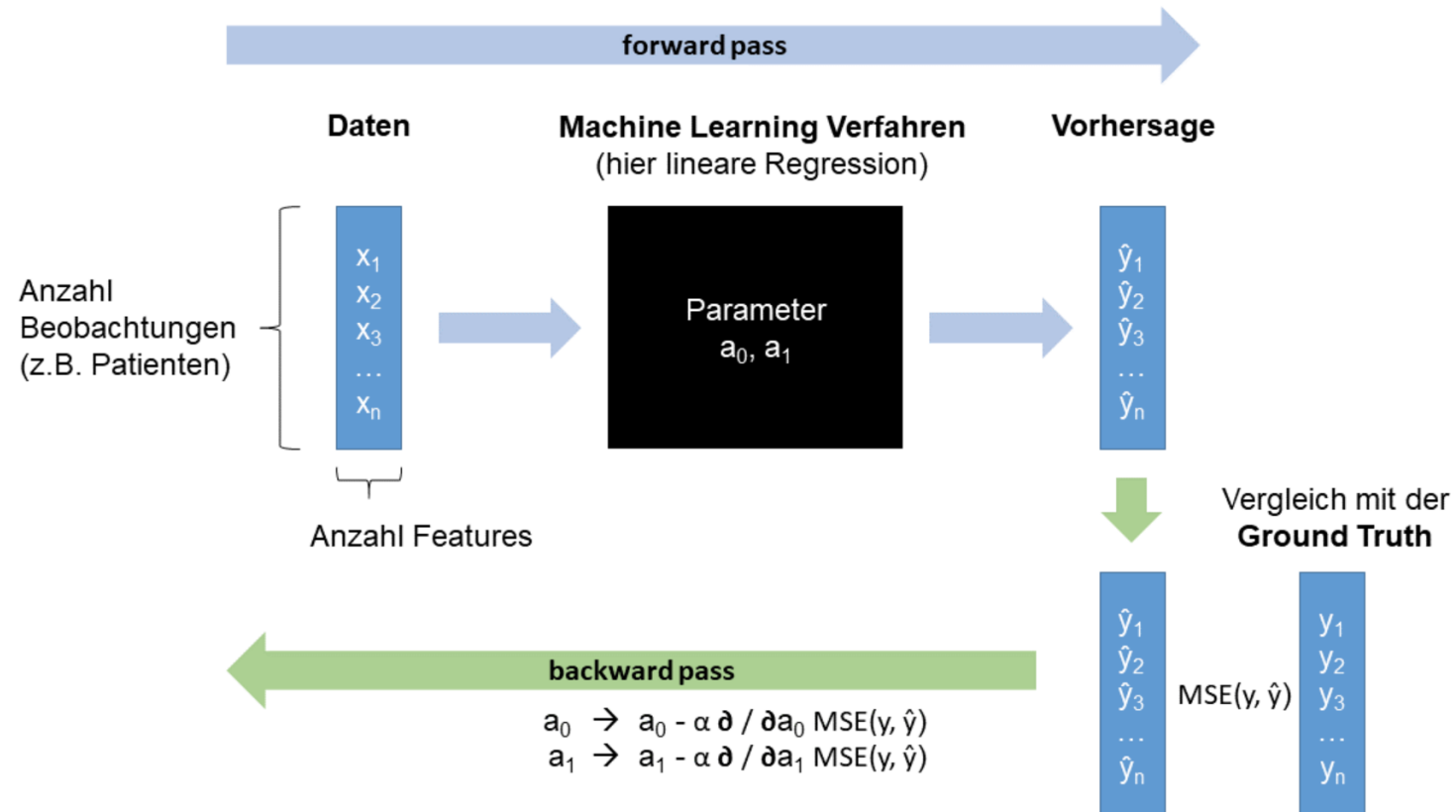
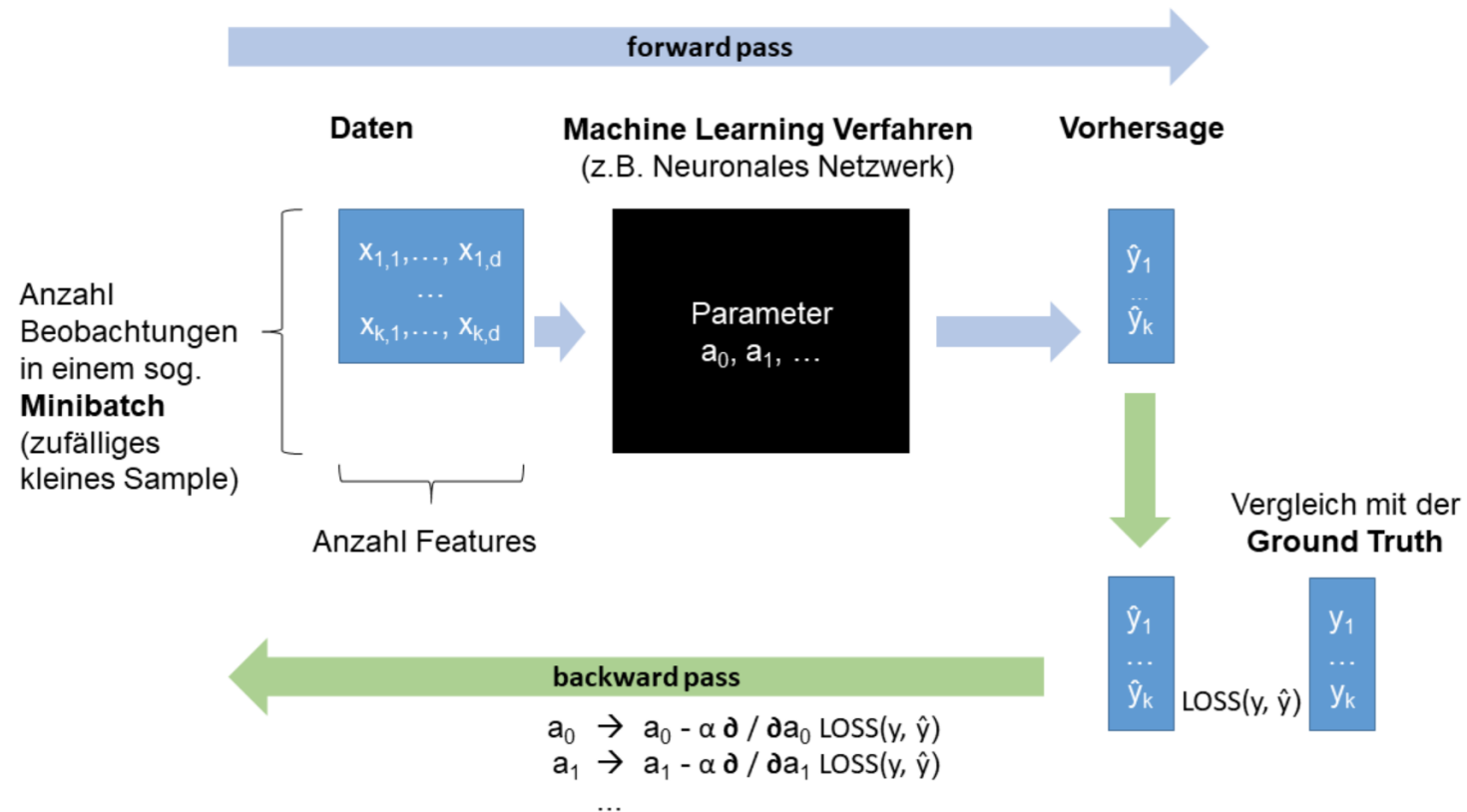


Schematisch für eine Lineare Regression $\hat{y}_i = a_0 + a_1 \cdot x_i$:



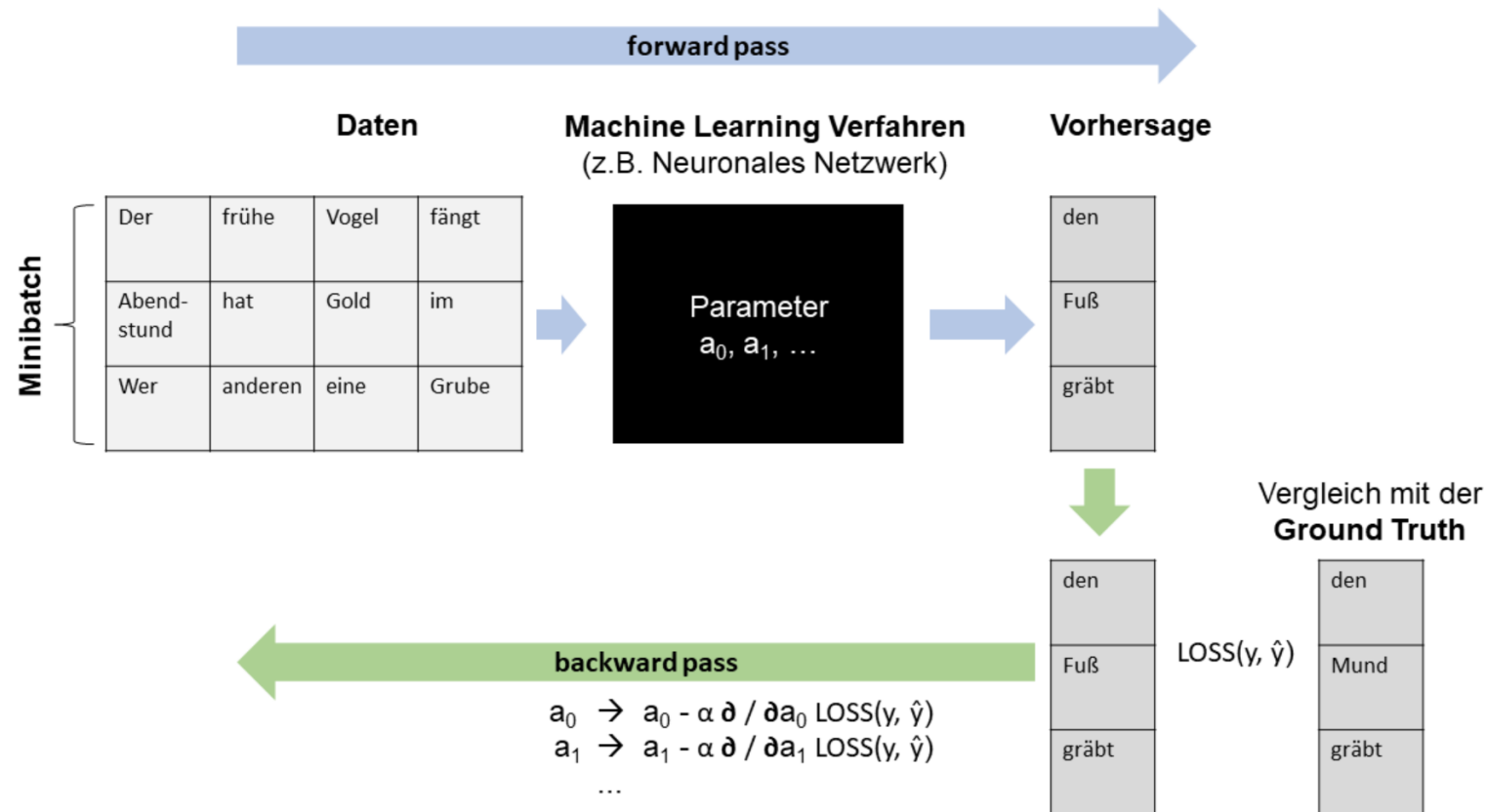


Das Machine Learning Verfahren könnte genauso ein **Neurales Netzwerk** sein
(und bei geeigneter Loss-Funktion auch eine Klassifikation statt Regression):



Frage: Wissen Sie, wie ein Large Language Model trainiert wird? (Was ist die Ground Truth?)

Training eines Large Language Models: Rate das nächste Wort.

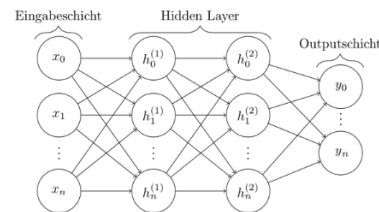


Vorteil des "self-supervised learnings": Supervised Learning mit gelabelten Daten ist möglich, ohne dass ein Mensch Labeling-Arbeit verrichten muss. Daher ist ein Training mit (gigantisch) großen Mengen an Trainingsdaten möglich!

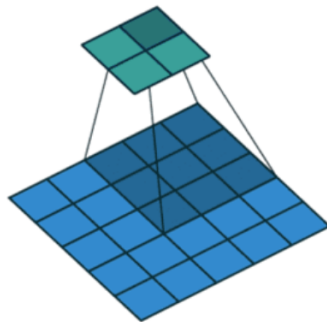
Netzwerk-Architekturen

Typische Architektur-Elemente:

- "Fully connected" für unstrukturierte Daten



- "Convolutional layer" für Bilddaten



https://www.topbots.com/intuitively-understanding-convolutions-deep-learning/stride_2_convolution_300px_web/

Für Text:

- Bis 2017: Rekurrente Neuronale Netzwerke (LSTM, GRU)
- Seit 2017: Transformer

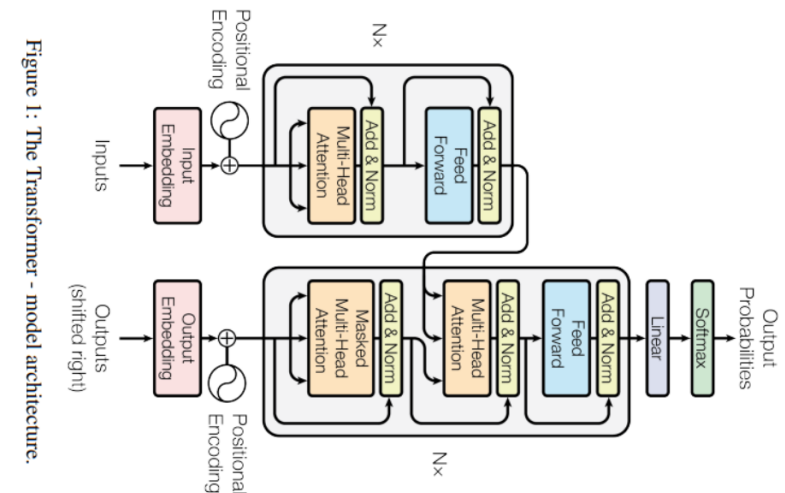


Figure 1: The Transformer - model architecture.

Vaswani, Ashish; Shazeer, Noam; Parmar, Niki; Uszkoreit, Jakob; Jones, Llion; Gomez, Aidan N.; Kaiser, Lukasz; Polosukhin, Illia (2017-06-12), "Attention Is All You Need". <https://arxiv.org/pdf/1706.03762.pdf>

Codierung von Wörtern in Vektoren: Verwende **Embeddings**, z.B. (0 0 1 1 1 0 0). Jedes Wort (genauer: "Token") wird durch einen Vektor im $\mathbb{R}^n$ repräsentiert, OHNE die Einschränkung, dass es sich um einen Einheitsvektor handeln muss.

Die Vektorenlänge kann je nach verfügbarer Rechenkapazität gewählt werden (z.B. 512).

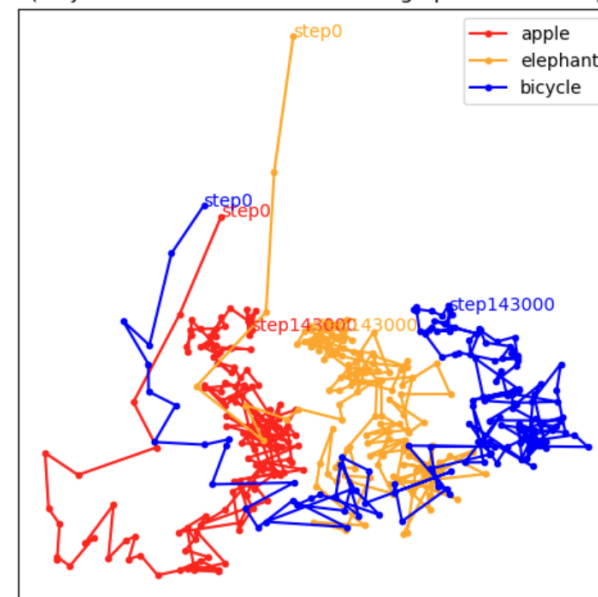
Embeddings werden mittrainiert.

Embeddings können räumlich interpretiert werden.

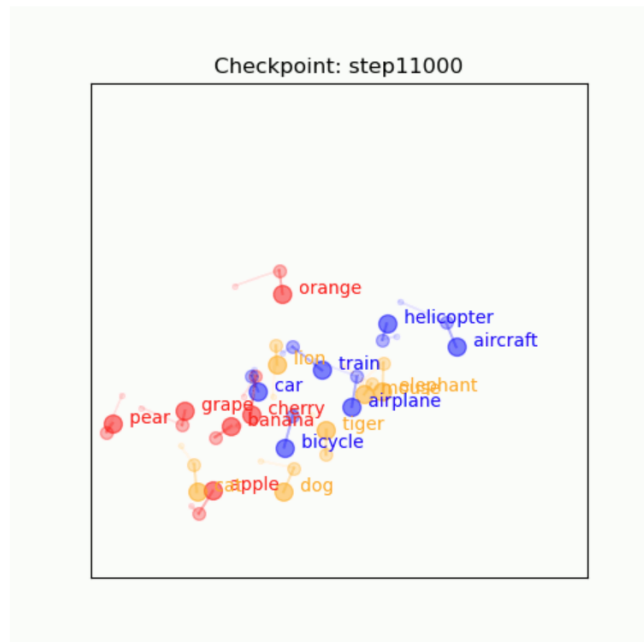
Visualisierung des Trainingsverlaufs mittels PCA:

Die PCA wurde auf dem letzten Trainingsschritt trainiert und bei allen vorherigen Schritten nur angewendet (um Konsistenz über verschiedene Zeitschritte zu erzielen); außerdem wurde die PCA nicht nur auf den 18 Begriffen angewendet, sondern auf der gesamten Embeddingmatrix, um die ausgewählten Embeddings im Kontext aller Embeddings zu sehen.

Veränderung der Embeddings während des Trainings
(Projektion des 512-dim Embedding Space auf 2-dim)



Embeddings verschiedener Begriffe im Trainingsverlauf (projiziert mittels PCA auf 2D):



Verwendung der Pythia-Modellfamilie:

In den Pythia-Modellen wurden Modelle verschiedener GröÙer auf die exakt gleiche Art und Weise (gleiche Hyperparameter, gleiche Trainingsdaten in der gleichen Reihenfolge, ...) trainiert und Zwischenstände des Trainings gespeichert.

Model: Checkpoint:	14m	31m	70m	160m	410m	1b	1.4b	2.8b	6.9b	12b
0	-	-	-	-	-	-	-	-	-	-
10000	-	-	-	-	-	-	-	-	-	-
20000	-	-	-	-	-	-	-	-	-	-
...	-	-	-	-	-	-	-	-	-	-
143000	-	-	-	-	-	-	-	-	-	-

```
In [49]: # source: https://github.com/EleutherAI/pythia
from transformers import GPTNeoXForCausalLM, \
    AutoTokenizer

model = GPTNeoXForCausalLM.from_pretrained(
    "EleutherAI/pythia-70m-deduped",
    revision="step143000",
    cache_dir="./pythia-70m-deduped/step143000",
)

tokenizer = AutoTokenizer.from_pretrained(
    "EleutherAI/pythia-70m-deduped",
    revision="step143000",
    cache_dir="./pythia-70m-deduped/step143000",
)

inputs = tokenizer("Hello, I am", return_tensors="pt")
tokens = model.generate(**inputs)
tokenizer.decode(tokens[0])

Setting `pad_token_id` to `eos_token_id`:0 f
or open-end generation.
```

```
Out[49]: 'Hello, I am looking for a way to get my name in the mai
1. I am looking for'
```

Frage: Wann lernt ein Modell gewisse Fähigkeiten? Welche Modellgröße ist hierfür notwendig?

Versuche *one-shot-learning*: `prompt = "one plus three is four.\n\nnone plus one is"`

Ergebnisse z.B. für pythia-70m:

- checkpoint 1000: 'one plus three is four.\n\nnone plus one is the most common.\n\nThe most'
- checkpoint 50000: 'one plus three is four.\n\nnone plus one is four is four is four is four is'
- checkpoint 143000: 'one plus three is four.\n\nnone plus one is<|endoftext|>'

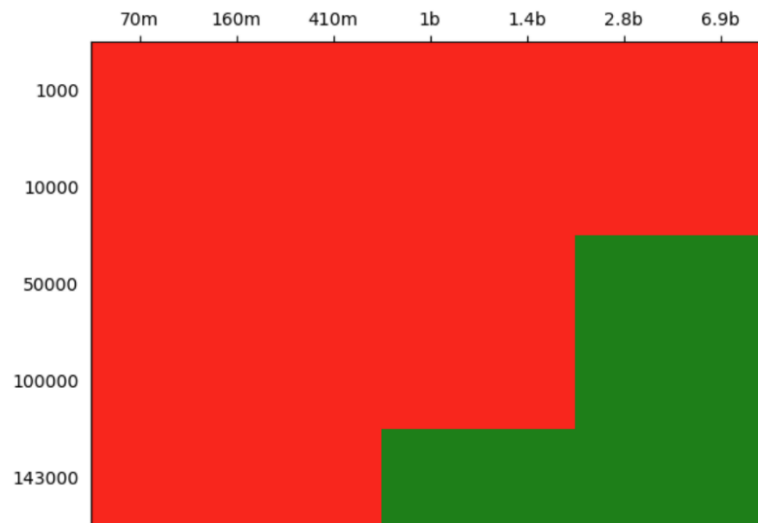
Ergebnisse z.B. für pythia-1b:

- checkpoint 1000: 'one plus three is four.\n\nnone plus one is four.\n\nnone plus one is'
- checkpoint 50000: 'one plus three is four.\n\nnone plus one is five.\n\nnone plus two is'
- checkpoint 143000: 'one plus three is four.\n\nnone plus one is two.\n\nnone plus two is'

Frage: Wann lernt ein Modell gewisse Fähigkeiten? Welche Modellgröße ist hierfür notwendig?

Versuche *one-shot-learning*: prompt = "one plus three is four.\n\nnone plus one is"

Ergebnisse:

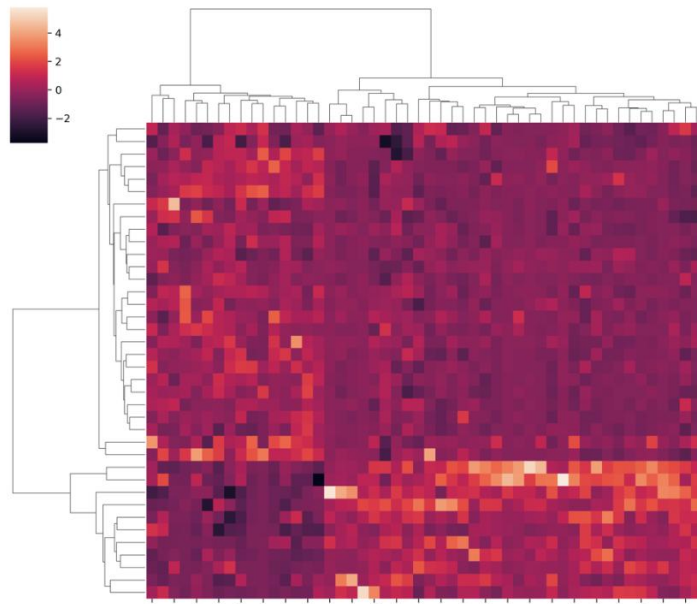


Motiviert durch das Versagen großer Sprachmodelle bei der "Alice in Wonderland"-Aufgabe:

"Alice hat N Brüder und M Schwestern. Wie viele Schwestern hat Alices Bruder?"



Marianna Nezhurina, Lucia Cipolina-Kun, Mehdi Cherti, Jenia Jitsev: "Alice in Wonderland: Simple Tasks Showing Complete Reasoning Breakdown in State-Of-the-Art Large Language Models." (<https://arxiv.org/abs/2406.02061>)



1. Wiederholung

2. Personalisierte Medizin

3. Clusterverfahren

```
In [4]: ▶ import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt  
  
import seaborn as sns
```

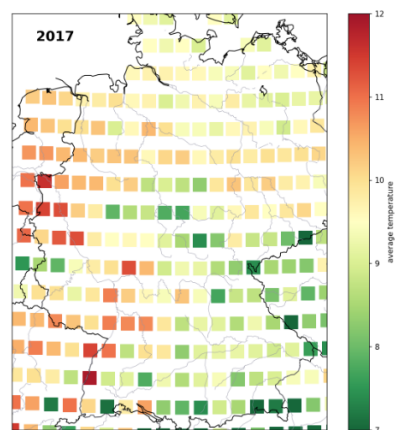
Wiederholung:

- Das Gesetz der großen Zahlen
- Der Zentrale Grenzwertsatz
- Holdout-Methode
- Wahrscheinlichkeit seltener Ereignisse
- Korrelation

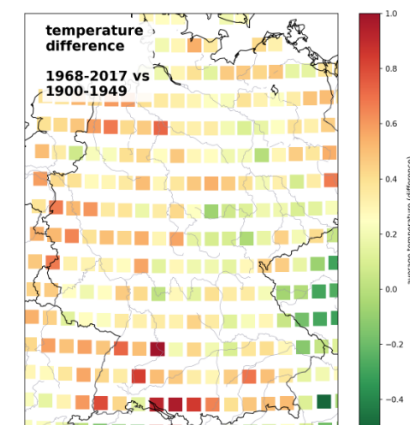
1.1 Rückblick

Wir haben Temperaturdaten eingelesen und ausgewertet:

Für 2017:



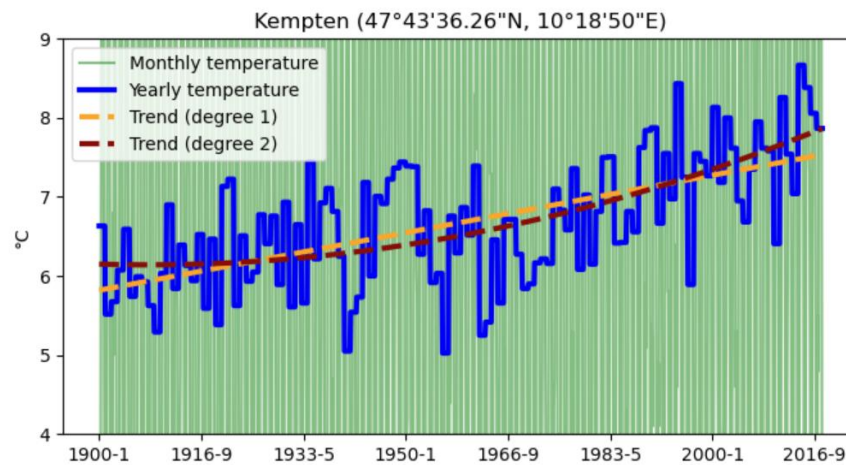
Visualisierung der relativen Erwärmung:



```
In [59]: ▶ # Temperaturdatensatz ausgewertet am Standort Kempten, Zoom
xvalues = list(range(len(df_temp["temperature"])))

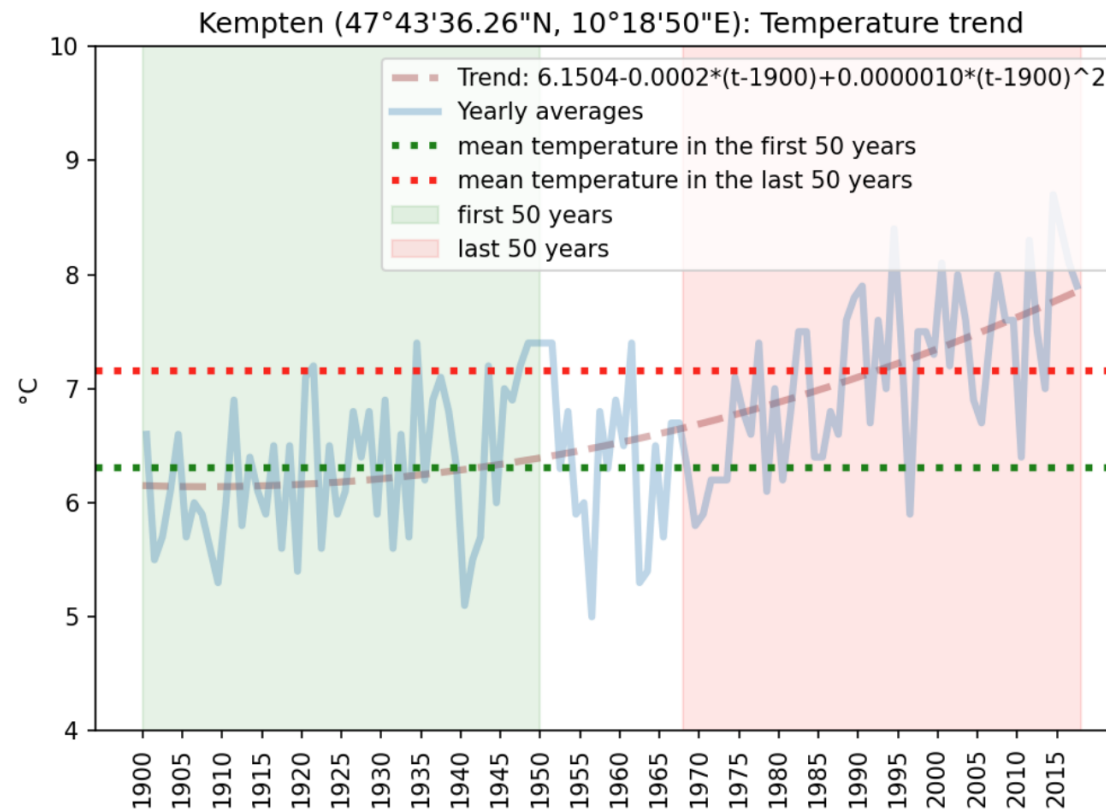
for deg in [1,2]: # calculate trends of degree 1 (linear) and 2 (quadratic)
    poly = np.poly1d ( np.polyfit( xvalues, df_temp["temperature"], deg=deg) )
    df_temp["trend_degree"+str(deg)] = poly ( xvalues )

fig,ax = plt.subplots( figsize=[8,4] )
df_temp["temperature" ].plot ( label="Monthly temperature", color="green", alpha=0.5 )
df_temp["mean" ].plot ( label="Yearly temperature", color="blue", linewidth=3 )
df_temp["trend_degree1"].plot ( label="Trend (degree 1)", color="orange", linewidth=3, linestyle="dashed" )
df_temp["trend_degree2"].plot ( label="Trend (degree 2)", color="darkred", linewidth=3, linestyle="dashed" )
ax.set_title ( "Kempten (47°43'36.26\"N, 10°18'50\"E)" ); ax.set_ylabel ("°C"); plt.legend(); plt.ylim(4,9);
```

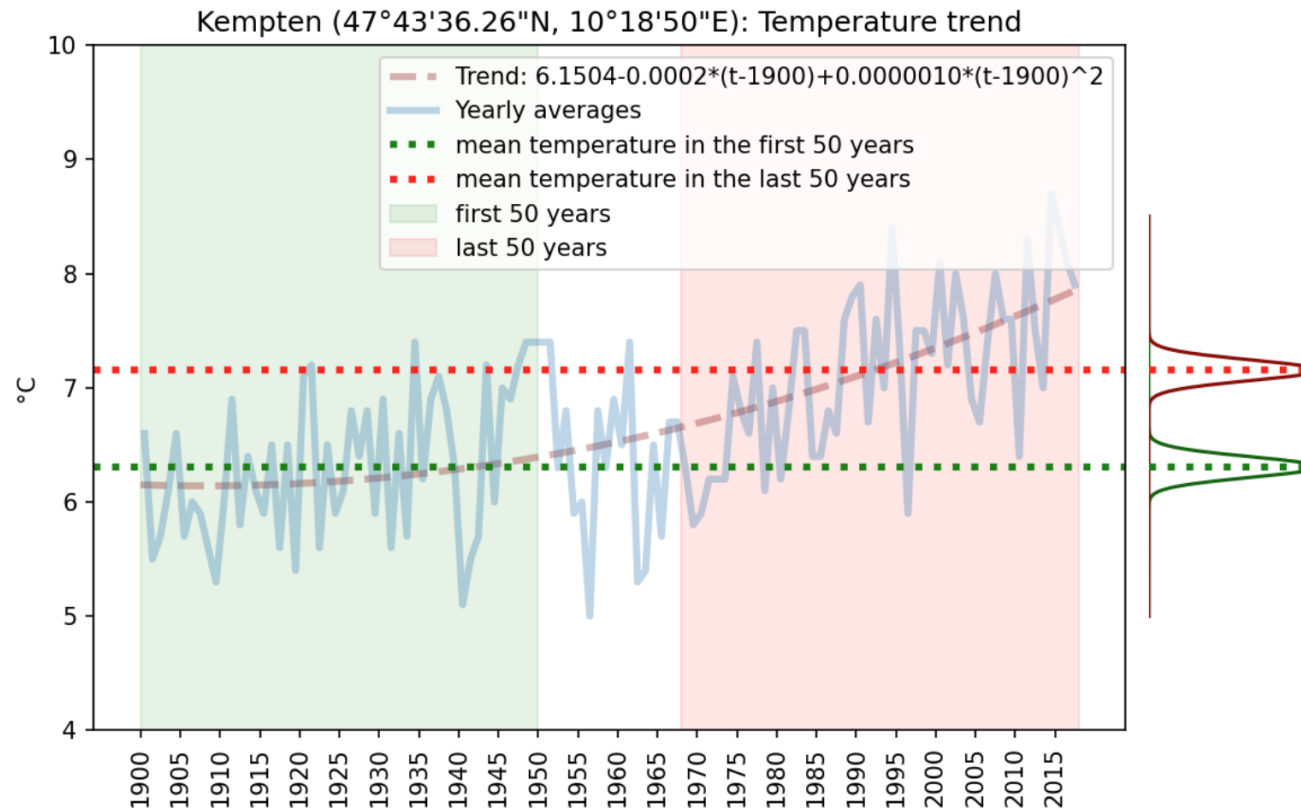


1.2 Gesetz der großen Zahlen und Zentraler Grenzwertsatz

Der Mittelwert vergleichbarer Jahre konvergiert wegen des **Gesetzes der großen Zahlen**:



Bei endlichem n (hier $n = 50$) fluktuiert der Mittelwert
(Zentraler Grenzwertsatz, Schwankungsbreite $\sim \text{const} \cdot \sqrt{1/n}$):



Fazit: Der zentrale Grenzwertsatz liefert eine Abschätzung darüber, wie weit der Mittelwert zufallsbedingt vom "wahren Wert" abweicht.
(Hier kann man also nicht behaupten, dass es in den letzten 50 Jahren nur zufällig wärmer aussieht.)

1.3 Ein Experiment: Gegenüberstellung mehrerer Datenquellen

Lese nun erneut die globalen Temperaturen (je Monat) der letzten Jahre ein:

```
In [7]: fpath = "VL10_Material/air_temp_2017/air_temp."
colnames = ["Longitude", "Latitude", "Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec", "Avg"]

res=[]
for year in range(2006,2018):
    dfw = pd.read_csv ( fpath+str(year), sep="\s+", header=None,
                        names=colnames ).drop("Avg",axis=1).set_index(["Longitude", "Latitude"]).T
    dfw.index = [str(year)+"-"+str(i+1) for i in range(len(dfw.index))]
    res.append(dfw)

df_weather = pd.concat(res,axis=0)
df_weather
```

```
Out[7]: Longitude -179.75 ... 179.75
Latitude 71.25 68.75 68.25 67.75 67.25 66.75 66.25 65.75 65.25 -16.75 ... -85.25 -85.75 -86.25 -86.75 -87.25 -87.75 -88.25 -88.75 -89.25 -89.75
2006-1 -22.9 -26.7 -28.0 -28.1 -32.0 -29.7 -25.8 -26.0 -27.3 27.2 ... -24.1 -22.9 -24.0 -25.1 -25.8 -26.3 -27.0 -27.2 -27.3 -27.4
2006-2 -21.6 -25.2 -25.9 -24.8 -27.6 -23.5 -18.4 -16.7 -17.0 27.3 ... -33.0 -32.9 -34.8 -36.4 -37.6 -38.7 -40.0 -40.8 -41.2 -41.6
2006-3 -24.1 -25.2 -25.8 -24.8 -27.2 -23.0 -18.1 -16.8 -16.3 28.0 ... -45.6 -45.9 -48.2 -50.1 -51.4 -52.7 -54.0 -54.8 -55.4 -55.8
2006-4 -19.9 -22.5 -23.4 -22.5 -25.1 -21.4 -17.0 -16.1 -16.7 27.9 ... -51.5 -51.2 -52.8 -53.9 -54.6 -55.2 -56.0 -56.2 -56.0 -55.8
2006-5 -5.5 -5.5 -6.4 -5.5 -8.0 -5.9 -3.7 -2.8 -2.4 26.7 ... -53.6 -53.0 -54.4 -55.4 -55.9 -56.1 -56.8 -56.9 -56.6 -56.1
...
2017-8 2.9 3.6 3.3 5.4 4.6 6.7 7.9 10.1 11.0 25.2 ... -52.2 -51.9 -53.6 -55.3 -56.5 -57.6 -58.9 -59.6 -59.8 -60.1
2017-9 0.7 0.1 -0.7 0.4 -1.6 0.8 2.8 4.6 5.5 25.6 ... -51.7 -51.5 -53.3 -54.7 -55.6 -56.4 -57.4 -57.8 -57.8 -57.8
2017-10 -3.1 -5.3 -6.7 -6.8 -10.3 -6.8 -2.7 -1.9 -2.6 26.8 ... -40.6 -40.4 -42.1 -43.5 -44.5 -45.2 -46.1 -46.4 -46.4 -46.5
2017-11 -6.7 -9.2 -10.7 -10.8 -14.4 -10.9 -6.6 -4.8 -4.6 27.1 ... -31.7 -30.7 -31.7 -33.0 -33.3 -34.0 -34.7 -35.0 -35.1 -35.6
2017-12 -13.3 -15.9 -17.1 -17.0 -20.7 -16.7 -11.2 -11.7 -14.0 28.6 ... -24.3 -23.1 -24.2 -25.2 -25.7 -26.3 -26.8 -27.0 -27.0 -27.2
```

144 rows × 85794 columns

Die Wetterdaten sollen nun einer anderen Datenquelle, dem Apple-Aktienkurs gegenübergestellt werden:

```
In [12]: # aus der Yahoo Finance API gezogen,
# als .csv zwischengespeichert
aapl = pd.read_csv("VL10_Material/aapl.csv")
aapl
```

Out[12]:

	Date	High	Low	Open	Close	Volume
0	2006-01-03	2.669643	2.580357	2.585000	2.669643	807234400.0
1	2006-01-04	2.713571	2.660714	2.683214	2.677500	619603600.0
2	2006-01-05	2.675000	2.633929	2.672500	2.656429	449422400.0
3	2006-01-06	2.739286	2.662500	2.687500	2.725000	704457600.0
4	2006-01-09	2.757143	2.705000	2.740357	2.716071	675040800.0
...	...	...	...	...	...	...
3015	2017-12-22	43.855000	43.625000	43.669998	43.752499	65397600.0
3016	2017-12-26	42.867500	42.419998	42.700001	42.642502	132742000.0
3017	2017-12-27	42.695000	42.427502	42.525002	42.650002	85992800.0
3018	2017-12-28	42.962502	42.619999	42.750000	42.770000	65920800.0
3019	2017-12-29	42.647499	42.305000	42.630001	42.307499	103999600.0

3020 rows × 9 columns

```
In [13]: # Verwende den Median je Monat als Zielgröße
stockprice = aapl.groupby(["year", "month"])\
["Close"].median()
stockprice
```

Out[13]:

```
year  month
2006   1      2.716785
      2      2.472143
      3      2.285357
      4      2.394286
      5      2.374285
      ...
2017   8     39.945000
      9     39.662500
     10     39.082500
     11     43.267502
     12     42.847500
Name: Close, Length: 144, dtype: float64
```

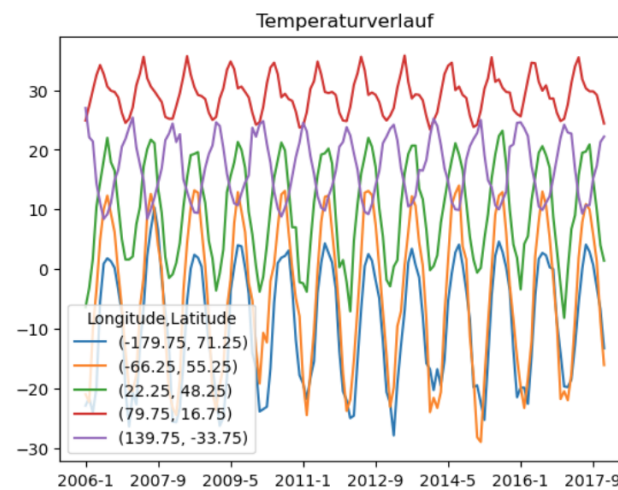
```
In [14]: # Finale Angleichung des Datenformats
stockprice.index = pd.Index([str(u)+"-"+str(v)
                             for (u,v) in stockprice.index])
stockprice
```

Out[14]:

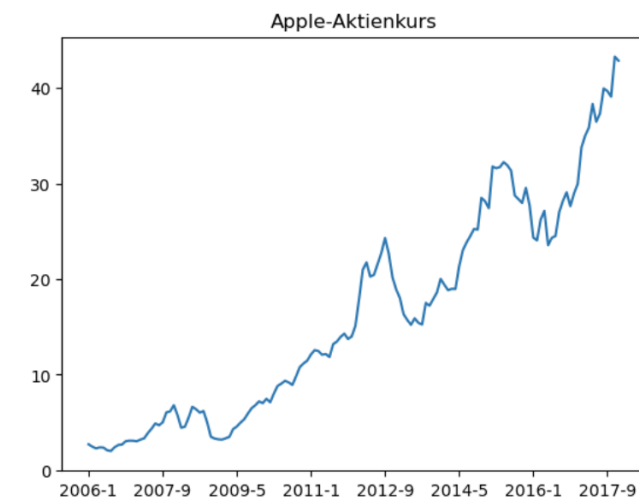
```
2006-1      2.716785
2006-2      2.472143
2006-3      2.285357
2006-4      2.394286
2006-5      2.374285
      ...
2017-8     39.945000
2017-9     39.662500
2017-10     39.082500
2017-11     43.267502
2017-12     42.847500
Name: Close, Length: 144, dtype: float64
```

Visualisierung der Temperaturverläufe an ausgewählten Wetterstationen und des Apple-Aktienkurses:

```
In [18]: df_weather.iloc[:,  
    [0, 20000, 40000, 60000, 80000] ].plot(  
    title = "Temperaturverlauf" );
```



```
In [17]: stockprice.plot( title="Apple-Aktienkurs" );
```

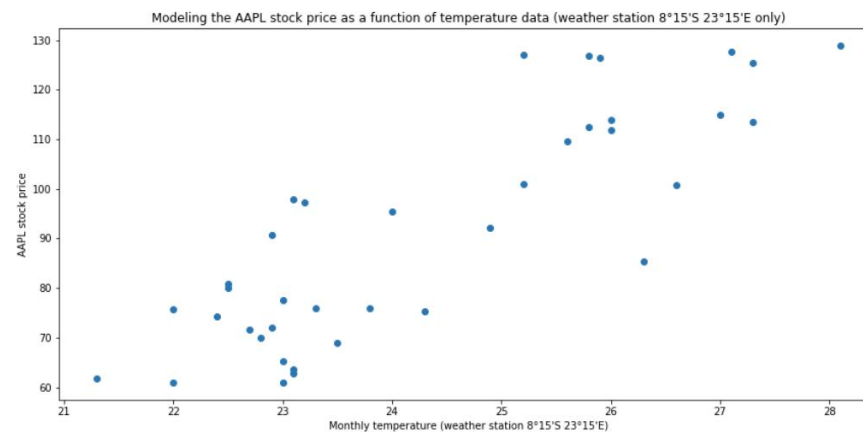


Frage: Können Sie ein Modell erstellen, das basierend auf den Temperaturdaten den Apple-Aktienkurs vorhersagt?

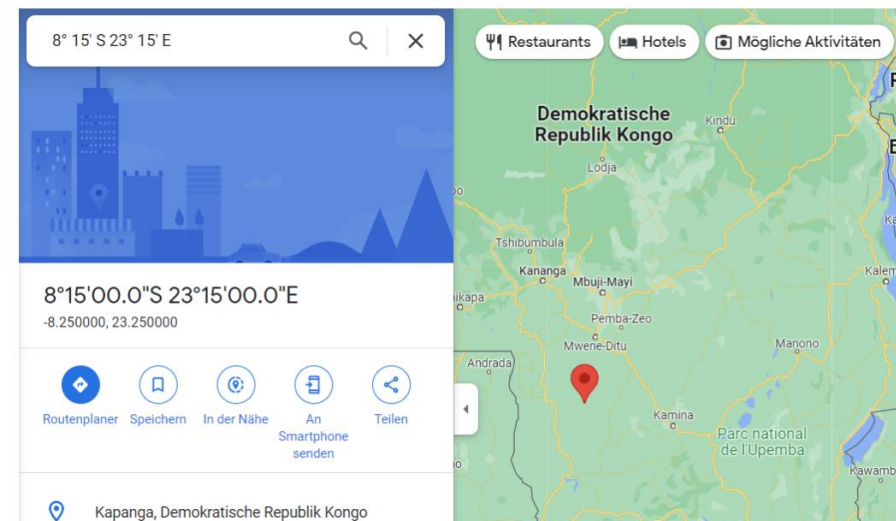
Antwort: Ja, der Aktienkurs kann gut vorhergesagt werden.

Hierbei wurde auf die **38 Monate** November 2012-Oktober 2015 und auf **6817** ausgewählte Wetterstationen eingegrenzt.

Es gibt sogar einzelne Wetterstationen mit hoher Vorhersagekraft:

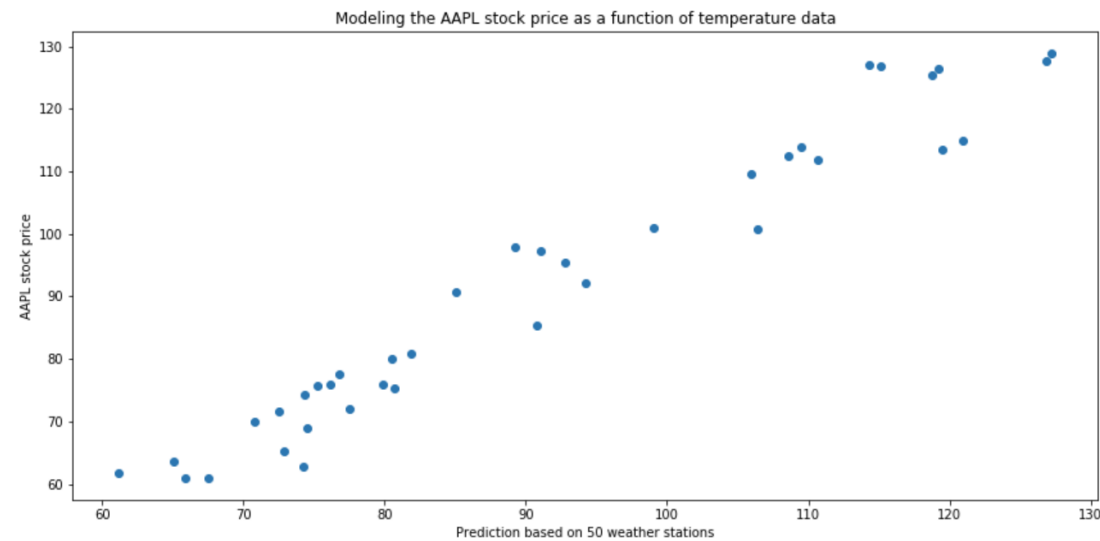


Wetterstation "8° 15' S 23° 15' E" zeigt eine 85%-Korrelation mit dem Apple-Aktienkurs.



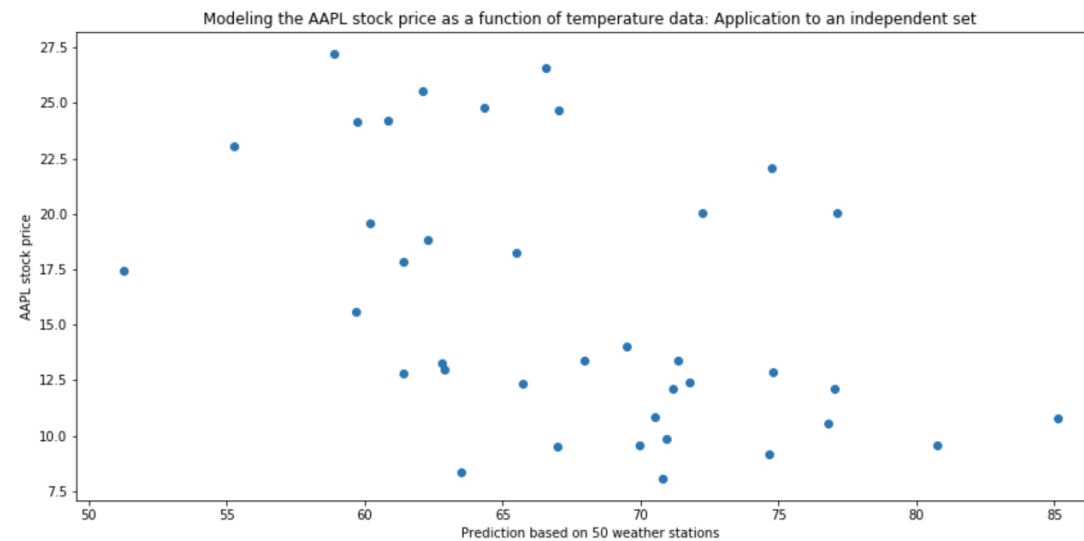
Die 50 besten Wetterstationen zusammen haben eine noch bessere Vorhersagekraft: Lineare Regression

$$\text{Apple-Aktienkurs} \approx a_0 + a_1 \text{Wetterstation}_1 + \dots + a_{50} \text{Wetterstation}_{50}$$



Frage: Haben wir ein gutes Modell zur Vorhersage von Aktienkursen gefunden?

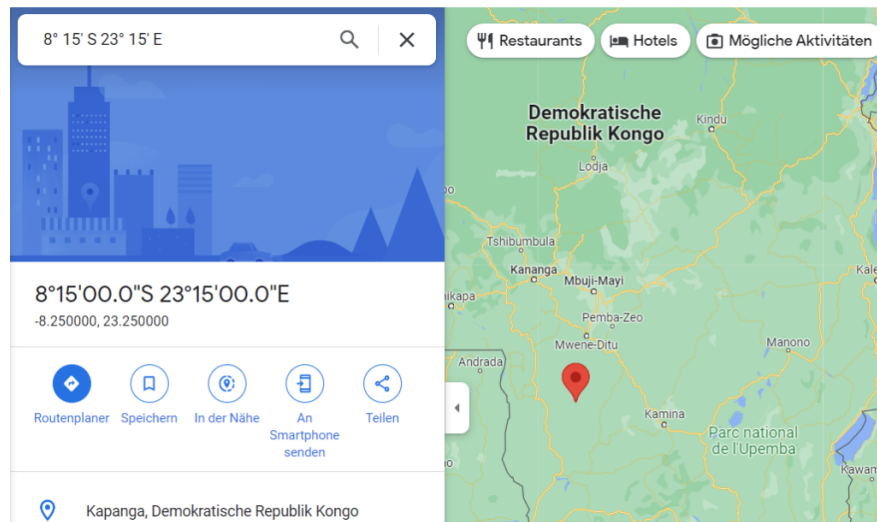
Überprüfung auf einem unabhängigen Datensatz (38 weitere Monate):



Nein, das Modell generalisiert nicht. (Holdout-Methode: Notwendigkeit eines Testdatensatzes!)

Zusammenfassung: Es wurde in sehr vielen Features nach Features mit hoher Vorhersagekraft gesucht. Einige Features sehen auf den Trainingsdaten vielversprechend aus. Die Überprüfung auf unabhängigen Daten zeigt, dass die Features allein aus Zufallsgründen ausgewählt wurden.

Wiederholung: Wahrscheinlichkeit für ein seltenes Ereignis (mit individueller Wahrscheinlichkeit 0,1%) bei häufigen Versuchen ($n=6817$):



$P(\text{mind. 1 seltenes Ereignis})$

$$= 1 - P(\text{genau 0 S.E.})$$

$$= 1 - \prod_{i=1}^n P(\text{im i.ten Versuch kein S.E.})$$

$$= 1 - (1 - 0.001)^{6817} = 99,89\%$$

1.4 Warum gerade diese 38 Monate?

Idee: Suche nach einem Zeitraum, in dem es hohe Korrelationen zwischen den Datenquellen gibt.

Berechnung der Korrelation in Numpy:

```
In [119]: ▶ # Berechnung der Korrelation mit np.corrcoef  
np.corrcoef ( [1,2,3,4,5],  
               [1,4,9,16,25] )
```

```
Out[119]: array([[1.          , 0.98110491],  
                 [0.98110491, 1.          ]])
```

```
In [79]: ▶ # Der eigentliche Korrelationskoeffizient  
# ist die Zahl "rechts oben"  
  
np.corrcoef ( [1,2,3,4,5],  
               [1,4,9,16,25] )[0,1]
```

```
Out[79]: 0.981104910251593
```

Auf unsere Daten angewendet:

```
In [83]: ▶ # Korrelation mit der ersten Wetterstation  
  
months = 38  
np.corrcoef(stockprice[:months],  
             df_weather.iloc[:months,0])
```

```
Out[83]: array([[1.          , 0.26522374],  
                 [0.26522374, 1.          ]])
```

```
In [82]: ▶ # Korrelation mit den ersten beiden Stationen  
# Frage: Wo kommt die 0.994 her?  
  
np.corrcoef(stockprice[:months],  
             df_weather.iloc[:months,:2].T)
```

```
Out[82]: array([[1.          , 0.26522374, 0.23302217],  
                 [0.26522374, 1.          , 0.99425536],  
                 [0.23302217, 0.99425536, 1.          ]])
```

Suche nach der höchsten Korrelation (suche 1. im größeren Zeitfenster nach guten 38 Monaten, 2. nach guten Wetterstationen):

```
In [70]: %%time

# Wir haben einen Suchzeitraum von 144 Monaten. In diesem suchen wir ein Zeitfenster von 38 Monaten
# sowie eine Wetterstation, für die die Korrelation besonders hoch ist.

cors=dict()
for offset in range(stockprice.shape[0]-months)::10:          # (wegen hoher Autokorrelation nur jede 10. Möglichkeit)
    print(offset)
    cors[offset]=[np.corrcoef ( stockprice[offset:offset+months],
                               df_weather.iloc[offset:offset+months,i] )[0,1] for i in range(df_weather.shape[1])]

print ( {k: np.max(cors[k]) for k in cors.keys()} )
# Ergebnis: Für offset="80 Monate" gibt es eine Wetterstation mit Korrelation 84,8%

0
10
20
30
40
50
60
70
80
90
100
{0: 0.582214004879787, 10: 0.644754835095684, 20: 0.7670327012044061, 30: 0.7674916459687929, 40: 0.6294255309802459, 50: 0.7163466776766
202, 60: 0.761504575584187, 70: 0.6929080346488516, 80: 0.8480475314072763, 90: 0.7077909000162199, 100: 0.471504183158686}
CPU times: total: 22min 1s
Wall time: 22min 20s
```

Die gefundene Wetterstation ist "8° 15' S 23° 15' E".
(Ähnlich können wir die 6817 "besten" Wetterstationen für die weitere Analyse finden.)

Fazit:

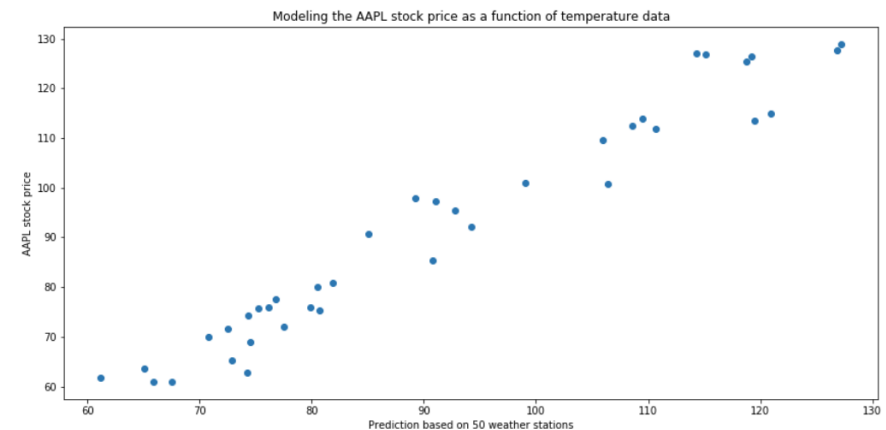
- Durch viele Features (=Spalten) und wenige Observationen (Monate oder Patienten, =Zeilen) fällt es leicht, aufsehenserregende Ergebnisse zu publizieren.
- Da der eigentliche Suchraum "144 Monate, 85794 Wetterstationen" noch wesentlich größer ist als die publizierten "38 Monate, 6817 Wetterstationen", sind seltene Ereignisse noch wahrscheinlicher.
- **Das Weglassen dieser Information ist aktives Schummeln!**

("Passives Schummeln" ist schwer zu vermeiden: Missglückte Experimente werden nicht publiziert; viele Gruppen arbeiten parallel, die schnellste publiziert; ...)

1.5 Warum gerade 6817 Wetterstationen?

Erinnerung: Wir konnten den Apple-Aktienkurs vorhersagen basierend auf ...

- 38 Monate
- 6817 Wetterstationen
- ... von denen die 50 "besten" ausgewählt wurden



$$\text{Apple-Aktienkurs} \approx a_0 + \sum_{i=1}^{50} a_i \text{Wetterstation}_i$$

In einer sehr bekannten Veröffentlichung wurde mit einem Datensatz der Größe 38x6817 Krebs-Varianten basierend auf 50 ausgewählten Features klassifiziert.

Golub et al. 1999

#Gene (insg.): 6817
#Gene (verwendet): 50
#Training Samples: 38
#Test Samples: 34

Molecular Classification of Cancer: Class Discovery and Class Prediction by Gene Expression Monitoring

T. R. Golub,^{1,2*} D. K. Slonim,¹ P. Tamayo,¹ C. Huard,¹ M. Gaasenbeek,¹ J. P. Mesirov,¹ H. Collier,¹ M. L. Loh,² J. R. Downing,³ M. A. Caligiuri,⁴ C. D. Bloomfield,⁴ E. S. Lander^{1,3*}

Although cancer classification has improved over the past 30 years, there has been no general approach for identifying new cancer classes (class discovery) or for assigning tumors to known classes (class prediction). Here, a generic approach to cancer classification based on gene expression monitoring by DNA microarrays is described and applied to human acute leukemias as a test case. A class discovery procedure automatically discovered the distinction between acute myeloid leukemia (AML) and acute lymphoblastic leukemia (ALL) without previous knowledge of these classes. An automatically derived class predictor was able to determine the class of new leukemia cases. The results demonstrate the feasibility of cancer classification based solely on gene expression monitoring and suggest a general strategy for discovering and predicting cancer classes for other types of cancer, independent of previous biological knowledge.

Molecular Classification of Cancer: Class Discovery and Class Prediction by Gene Expression
Science 286:531-537. (1999). Published: 1999.10.14
T.R. Golub, D.K. Slonim, P. Tamayo, C. Huard, M. Gaasenbeek, J.P. Mesirov, H. Collier, M. Loh, J.R. Downing, M.A. Caligiuri, C.D. Bloomfield, and E.S. Lander
The microarray data is available via <https://www.kaggle.com/crawford/gene-expression>

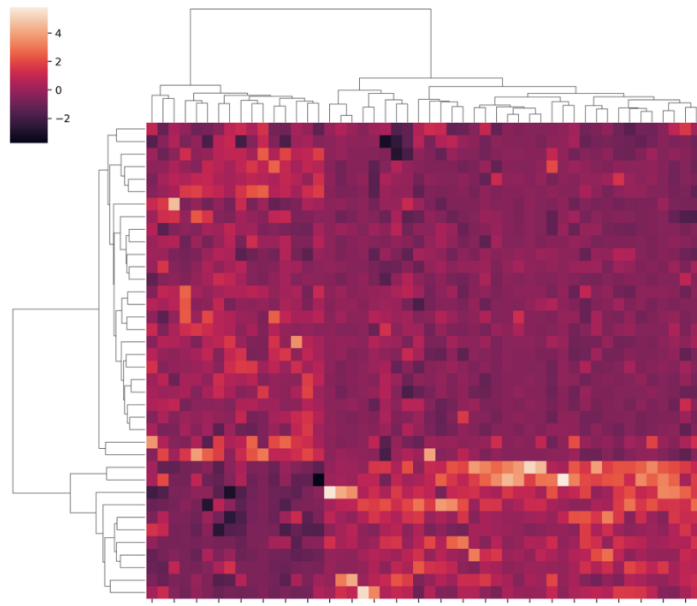
Our initial leukemia data set consisted of 38 bone marrow samples (27 ALL, 11 AML) obtained from acute leukemia patients at the time of diagnosis (13). RNA prepared from bone marrow mononuclear cells was hybridized to high-density oligonucleotide microarrays, produced by Affymetrix and containing probes for 6817 human genes (14). For each gene, we obtained a quantitative expression level. Samples were subjected to a priori

We applied this approach to the 38 acute leukemia samples. The set of informative genes to be used in the predictor was chosen to be the 50 genes most closely correlated with AML-ALL distinction in the known

We then created a 50-gene predictor on the basis of all 38 samples and applied it to an independent collection of 34 leukemia samples. The specimens consisted of 24 bone

Glauben wir das?

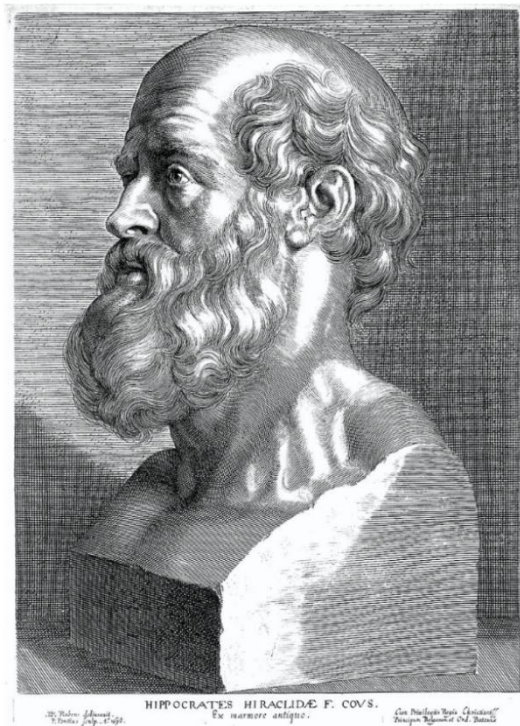
Fazit: Skepsis ist angebracht, wenn basierend auf vielen Features, aber wenigen Beobachtungen eine Aussage gemacht wird. (Bei Golup et al. wurde das Modell immerhin auf einem "unabhängigen" Testdatensatz evaluiert.)



1. Wiederholung

2. Personalisierte Medizin

3. Clusterverfahren



<https://en.wikipedia.org/wiki/Hippocrates>

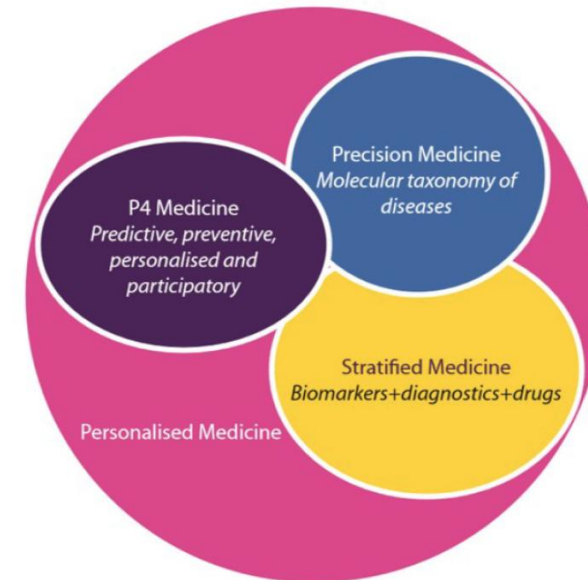
"Es ist wichtiger zu wissen, welche Person eine Krankheit hat, als zu wissen, welche Krankheit eine Person hat."

Hippokrates

2.1 Personalisierte Medizin

Ähnlich verwendete Begriffe:

- Personalised Medicine
- Precision Medicine
- Stratified Medicine
- P4 Medicine
 - predictive
 - preventive
 - personalised
 - participatory



**Many names for one concept
or many concepts in one name?**

PHG Foundation (<https://www.phgfoundation.org/>)

PHG Foundation is the working name of the Foundation for Genomics and Population Health. A charitable company registered in England and Wales.

The Personalised Medicine Technology Landscape:

Microbiome analysis	Epigenomics	3D imaging and printing	Consumer m-health apps	Wearables and sensors
Metabolomics	Proteomics	Genome editing /therapy	Implantable biosensors	Point of care testing devices
ctDNA	Single cell 'omics	Stem cell therapy	EPR dependent technologies	Microfluidics
Pathogen Genomics	Transcriptomics	Robotics	Internet of things	Synthetic biology
Genomics	Pharmaco-genomics	Virtual and augmented reality	Machine learning	Nanomedicine

- Technologies for greater molecular level characterisation
- Technologies for personalised therapeutic interventions
- Technologies for personalised disease and health monitoring
- Underpinning and enabling technologies

"The personalised medicine technology landscape", S.22
<https://www.phgfoundation.org/report/personalised-medicine-technology-landscape>

Definition "Personalised Medicine" of the European Commission (Horizon 2020 Advisory Group):

a medical model using characterisation of individuals' phenotypes and genotypes

(e.g. molecular profiling, medical imaging, lifestyle data)

for tailoring the right therapeutic strategy for the right person at the right time,

and/or to determine the predisposition to disease and/or to deliver timely and targeted prevention."

https://ec.europa.eu/info/research-and-innovation/research-area/health-research-and-innovation/personalised-medicine_en

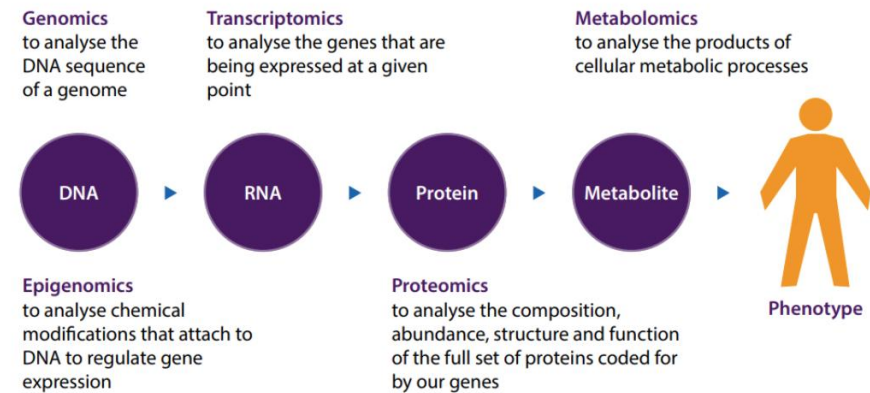
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Begriffsklärung "Omics":



"The personalised medicine technology landscape", S.40
<https://www.phgfoundation.org/report/personalised-medicine-technology-landscape>

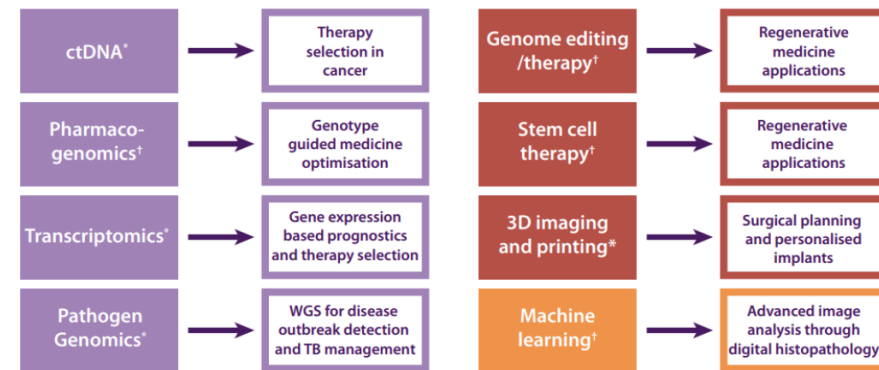
The Personalised Medicine Technology Landscape (long list):

Microbiome analysis	Epigenomics	3D imaging and printing	Consumer m-health apps	Wearables and sensors
Metabolomics	Proteomics	Genome editing /therapy	Implantable biosensors	Point of care testing devices
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"The personalised medicine technology landscape", S.22
<https://www.phgfoundation.org/report/personalised-medicine-technology-landscape>

The Personalised Medicine Technology Landscape (short list):



"The personalised medicine technology landscape", S.23
<https://www.phgfoundation.org/report/personalised-medicine-technology-landscape>

2.2 Ein Blick in die Anfangsjahre der Personalisierten Medizin

Golub et al. 1999

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#Training Samples: 38
#Test Samples: 34

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The microarray data is available via <https://www.kaggle.com/crawford/gene-expression>

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Although cancer classification has improved over the past 30 years, there has been no general approach for identifying new cancer classes (class discovery) or for assigning tumors to known classes (class prediction). Here, a generic approach to cancer classification based on gene expression monitoring by DNA microarrays is described and applied to human acute leukemias as a test case. A class discovery procedure automatically discovered the distinction between acute myeloid leukemia (AML) and acute lymphoblastic leukemia (ALL) without previous knowledge of these classes. An automatically derived class predictor was able to determine the class of new leukemia cases. The results demonstrate the feasibility of cancer classification based solely on gene expression monitoring and suggest a general strategy for discovering and predicting cancer classes for other types of cancer, independent of previous biological knowledge.

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We then created a 50-gene predictor on the basis of all 38 samples and applied it to an independent collection of 34 leukemia samples. The specimens consisted of 24 bone

Van 't Veer et al. 2002

#Gene (insg.): 25000
#Gene (verwendet): 70
#Training Samples: 78 (98)
#Test Samples: 19

Gene expression profiling predicts clinical outcome of breast cancer

Nature. 415 (6871): 530-6. doi:10.1038/415530a. hdl:1874/15552. PMID 11823860.

van 't Veer LJ, Dai H, van de Vijver MJ, et al. (2002).

The microarray data is available via Bioconductor: <https://www.bioconductor.org/packages/release/data/experiment/html/mammaPrintData.html>

Gene expression profiling predicts clinical outcome of breast cancer

Laura J. van 't Veer[†], Hongyue Dai[‡], Marc J. van de Vijver[†], Yudong D. He[†], Augustinus A. M. Hart[†], Mao Mao[†], Hans L. Peterse[†], Karin van der Kooy[†], Matthew J. Marton[†], Anke T. Witteveen[†], George J. Schreiber[†], Ron M. Kerkhoven[†], Chris Roberts[†], Peter S. Linsley[†], René Bernards[†] & Stephen H. Friend[†]

^{*} Divisions of Diagnostic Oncology, Radiotherapy and Molecular Carcinogenesis and Center for Biomedical Genetics, The Netherlands Cancer Institute, 121 Plesmanlaan, 1066 CX Amsterdam, The Netherlands
[†] Rosetta Inpharmatics, 12040 115th Avenue NE, Kirkland, Washington 98034, USA

[‡] These authors contributed equally to this work

Breast cancer patients with the same stage of disease can have markedly different treatment responses and overall outcome. The strongest predictors for metastases (for example, lymph node status and histological grade) fail to classify accurately breast tumours according to their clinical behaviour¹⁻³. Chemotherapy or hormonal therapy reduces the risk of distant metastases by approximately one-third; however, 70-80% of patients receiving this treatment would have survived without it^{4,5}. None of the signatures of breast cancer gene expression reported to date⁶⁻¹² allow for patient-tailored therapy strategies. Here we used DNA

carcinomas. Two hybridizations were carried out for each tumour using a fluorescent dye reversal technique on microarrays containing approximately 25,000 human genes synthesized by inkjet technology¹³. Fluorescence intensities of scanned images were

expression levels of the remaining samples from the good and the poor prognosis patients, respectively. The accuracy improved until the optimal number of marker genes was reached (70 genes).

The expression pattern of the 70 genes in the 78 samples is shown in the colour plot of Fig. 2b (left panel), where tumours were

The 78 sporadic lymph-node-negative patients were selected specifically to search for a prognostic signature in their gene expression profiles. Forty-four patients remained free of disease

To validate the prognosis classifier, an additional independent set of primary tumours from 19 young, lymph-node-negative breast cancer patients was selected. This group consisted of 7 patients who

Evaluation of Genomic Applications in Practice and Prevention (EGAPP)

The EGAPP was launched by the CDC Office of Public Health Genomics in the fall of 2004.

The initiative's goal is to establish and evaluate a systematic, evidence-based process for assessing genetic tests (...) in transition from research to clinical and public health practice.

<https://www.cdc.gov/egappreviews/recommendations/index.html>
<https://www.nature.com/articles/gim20092>
<https://www.nature.com/articles/gim2013110>

Nachträgliche Bewertung einiger Testverfahren:

Disease/Condition	Genetic Test/Purpose	Test Use	EGAPP Recommendation
Breast Cancer:	Can tumor gene expression profiling improve outcomes in patients with breast cancer?	Prognostic: association of genetic markers with disease severity, progression	Insufficient evidence to recommend for or against use Read the EGAPP recommendation (January 2006)
	Does the use of Oncotype DX tumor gene expression profiling to guide treatment decisions improve outcomes in patients with breast cancer?	Prognostic: association of genetic markers with disease severity, progression	Insufficient evidence to recommend for or against use Read the EGAPP recommendation (December 2015)
Cardiovascular Disease:	Recommendations from the EGAPP Working Group: Genomic profiling to assess cardiovascular risk to improve cardiovascular health	Prediction: association of genetic markers with disease risk	Insufficient evidence to recommend for or against use Read the EGAPP recommendation (December 2016)
	Genetic testing for Factor V Leiden and prothrombinase variants for diagnosis, venous thromboembolism (VTE)	Diagnostic: association of a genetic marker with a disease/condition in a person with symptoms and family members without symptoms	Sufficient evidence to recommend against routine use in adults with idiopathic VTE, and in their adult family members with no history or symptoms of thrombotic events, for medical decision making on preventive use of anticoagulants Read the EGAPP recommendation (January 2011)
Colorectal Cancer:	Can testing of tumor tissue for mutations in EGFR pathway downstream effector genes in patients with metastatic colorectal cancer improve health outcomes by guiding decisions regarding anti-EGFR therapy?	Pharmacogenomic: association of a genetic marker with a particular treatment in a person with the specified disease.	Sufficient evidence to recommend clinical use of KRAS mutation analysis to determine which patients are KRAS mutation positive and therefore unlikely to benefit from these agents before initiation of therapy. Insufficient evidence to recommend for or against BRAF V600E testing for the same clinical scenario. Insufficient evidence to recommend for or against testing for mutations in NRAS, or PIKCA, and/or loss of expression of PTEN or AKT proteins. Read the EGAPP recommendation (February 2015)

	Genetic testing strategies in newly diagnosed individuals with colorectal cancer aimed at reducing morbidity and mortality from Lynch syndrome in relatives	Diagnostic: association of a genetic marker with a disease/condition in a person with symptoms	Sufficient evidence to recommend use for the benefit of relatives Read the EGAPP recommendation (January 2009)
	Can CYP2A7 genotyping reduce morbidity and mortality in patients with rheumatoid arthritis?	Pharmacogenomic: predicting response to drug treatment	Insufficient evidence to recommend for or against use Read the EGAPP recommendation (January 2009)
Depression:	Testing for cytochrome P450 polymorphisms in adults with major depressive disorder treated with selective serotonin reuptake inhibitors	Pharmacogenomic: predicting response to drug treatment	Insufficient evidence to recommend for or against use, but use discouraged pending further research Read the EGAPP recommendation (December 2007)
Diabetes:	Does genomic profiling to assess Type 2 Diabetes (T2D) risk improve health outcomes?	Association of a genetic marker(s) with T2D risk profiling in general and high-risk populations.	*The EGAPP Working Group (EWG) found insufficient evidence to recommend testing for predictive variants in 28 variants (listed in Table 1) to assess risk for Type 2 Diabetes in the general population, based on studies in populations of northern European descent. *The EGAPP Working Group (EWG) found insufficient evidence to recommend testing for the TCF7L2 gene to assess risk for Type 2 Diabetes in high-risk individuals. Read the EGAPP recommendation (March 2013)
Prostate Cancer:	Does PCA3 Testing for the Diagnosis and Management of Prostate Cancer improve patient health outcomes?	Diagnostic and Prognostic: association of genetic markers with disease severity, progression	Insufficient evidence to recommend for or against use Read the EGAPP recommendation (September 2015)

<https://www.cdc.gov/egappreviews/recommendations/index.html>

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Cardiovascular Disease:	Recommendations from the EGAPP Working Group: Genetic profiling to assess cardiovascular risk to improve cardiovascular health	Predictive: association of genetic markers with disease risk	Insufficient evidence to recommend for or against use Read the EGAPP recommendation (2 December 2016)
	Genetic testing for Factor V Leiden and prothrombin variants for idiopathic venous thromboembolism (VTE)	Diagnostic: association of a genetic marker with a disease/condition in a person with symptoms and family members without symptoms	Sufficient evidence to recommend against routine use in adults with idiopathic VTE, and in their adult family members with no history or symptoms of thrombotic events, for medical decision making on preventive use of anticoagulants Read the EGAPP recommendation (2 January 2011)
Colorectal Cancer:	Can testing of tumor tissue for mutations in EGFR pathway downstream effector genes in patients with metastatic colorectal cancer improve health outcomes by guiding decisions regarding anti-EGFR therapy?	Pharmacogenomic: association of a genetic marker with a particular treatment in a person with the specified disease	Sufficient evidence to recommend clinical use of ERAs: mutation analysis to determine which patients are KRAS mutation positive and therefore unlikely to benefit from these agents before initiation of therapy Insufficient evidence to recommend for or against BRAF-V600E testing for the same clinical scenario Insufficient evidence to recommend for or against testing for mutations in NRAS, or PIK3CA, and/or loss of expression of PTEN or AKT proteins Read the EGAPP recommendation (2 February 2013)

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	Can LIG7A genotyping reduce morbidity and mortality in patients with metastatic colorectal cancer?	Pharmacogenomic: predicting response to drug treatment	Insufficient evidence to recommend for or against use Read the EGAPP recommendation (2 January 2009)
Depression:	Testing for cytochrome P450 polymorphisms in adults with major depressive disorders treated with selective serotonin reuptake inhibitors	Pharmacogenomic: predicting response to drug treatment	Insufficient evidence to recommend for or against use, but use discouraged pending further research Read the EGAPP recommendation (2 December 2007)
Diabetes:	Does genomic profiling to assess Type 2 Diabetes (T2D) risk improve health outcomes?	Association of a genetic marker with T2D risk profiling in general and high risk populations	*The EGAPP Working Group (IWG) found insufficient evidence to recommend testing for predictive variants in 28 variants (listed in Table 1) to assess risk for Type 2 Diabetes in the general population, based on studies in populations of northern European descent. *The EGAPP Working Group (IWG) found insufficient evidence to recommend testing for the TCF7L2 gene to assess risk for Type 2 Diabetes in high-risk individuals. Read the EGAPP recommendation (2 March 2013)
Prostate Cancer:	Does PCA3 Testing for the Diagnosis and Management of prostate cancer improve patient health outcomes?	Diagnostic and Prognostic: association of genetic markers with disease severity, progression	Insufficient evidence to recommend for or against use Read the EGAPP recommendation (2 September 2013)

<https://www.cdc.gov/egappreviews/recommendations/index.html>

Ergebnis bei vielen Testverfahren: "Insufficient evidence to recommend for or against use"

Erneut: Es ist schwierig, aus wenig Daten mit vielen Features valide Aussagen zu generieren.

Why Most Published Research Findings Are False.

Ioannidis J.P.A. (2005), PLoS Med 2(8): e124, <https://doi.org/10.1371/journal.pmed.0020124>; <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.0020124>

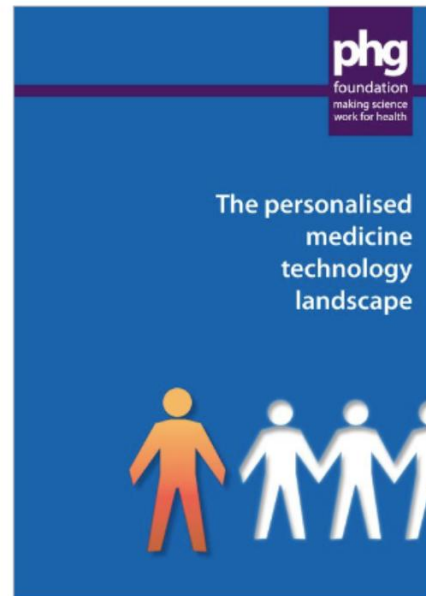
Corollary 1: *The smaller the studies conducted in a scientific field, the less likely the research findings are to be true.*

Corollary 3: *The greater the number (...) of tested relationships in a scientific field, the less likely the research findings are to be true.*

Thus, research findings are more likely true in confirmatory designs, such as large phase III randomized controlled trials (...) than in hypothesis-generating experiments.

[Findings in] fields considered highly informative and creative given the wealth of the assembled and tested information, such as microarrays and other high-throughput discovery-oriented research [4,8,17], should [be] extremely [less likely to be true].

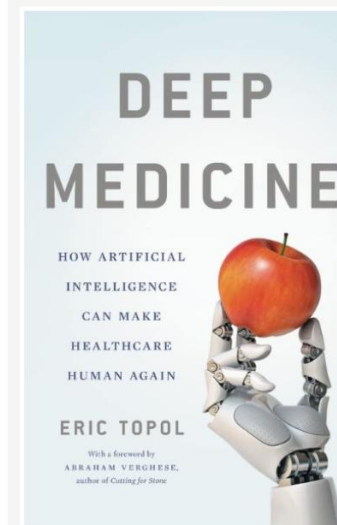
2.3 Further reading



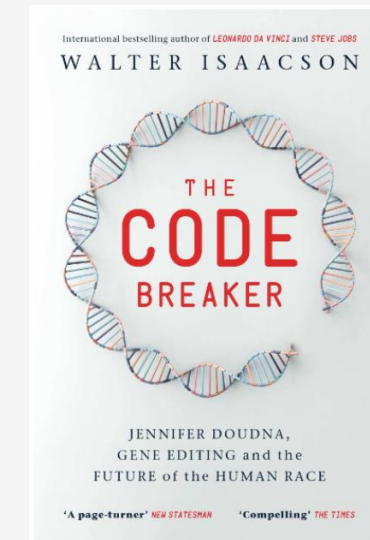
<https://www.phgfoundation.org/report/personalised-medicine-technology-landscape>

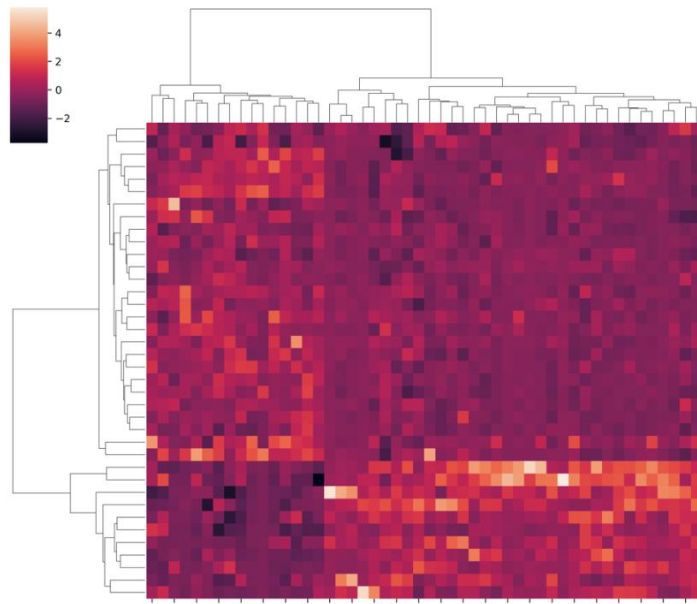
Weitere bekannte Bücher:

Ein breiter Blick auf KI-
Anwendungen in der Medizin:



Speziell auf die neuesten Entwicklungen
zu mRNA-Technologien:





1. Wiederholung

2. Personalisierte Medizin

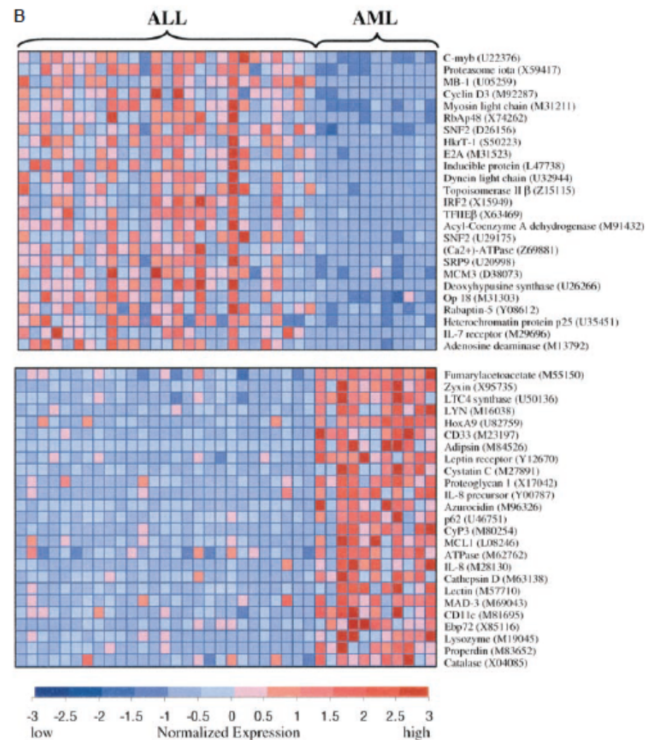
3. Clusterverfahren

Definition

Clustering is the task of grouping the objects based on similarity and dissimilarity.

-- 'Extensive Survey on Hierarchical Clustering Methods in Data Mining' (Dipak P Dabhi, Mihir R Patel)

3.1 Die Originaldaten aus dem Paper "Golub et al., 1999"



```
In [53]: df_train=pd.read_csv("Übungen/Golub1999/df_train.csv")\
          .set_index("patient")
          df_train.shape
```

Out[53]: (38, 7130)

Etwas skeptisch machen sollte, dass im Paper 6817, in den Originaldaten aber 7129 Gene erwähnt werden.

Die Zielgröße `actual` enthält zwei Leukämie-Varianten "ALL" oder "AML".

11x acute myeloid leukemia (AML)
27x acute lymphoblastic leukemia (ALL)

In [78]: ▶ df_train.head()

Out[78]:

	AFFX- BioB- 5_at	AFFX- BioB- M_at	AFFX- BioB- 3_at	AFFX- BioC- 5_at	AFFX- BioC- 3_at	AFFX- BioDn- 5_at	AFFX- BioDn- 3_at	AFFX- CreX- 5_at	AFFX- CreX- 3_at	AFFX- BioB- 5_st	...	U48730_at	U58516_at	U73738_at	X06956_at	X16699_at	X83863_at	Z17240_at	L49218_f_
patient																			
1	-214	-153	-58	88	-295	-558	199	-176	252	206	...	185	511	-125	389	-37	793	329	36
2	-139	-73	-1	283	-264	-400	-330	-168	101	74	...	169	837	-36	442	-17	782	295	11
3	-76	-49	-307	309	-376	-650	33	-367	206	-215	...	315	1199	33	168	52	1138	777	41
4	-135	-114	265	12	-419	-585	158	-253	49	31	...	240	835	218	174	-110	627	170	-50
5	-106	-125	-76	168	-230	-284	4	-122	70	252	...	156	649	57	504	-26	250	314	14

5 rows × 7129 columns

3.2 Auswahl der 50 "wichtigsten" Gene wie im Paper

In der Originalveröffentlichung wurde die Korrelation zwischen den Genexpressionswerten und der Zielgröße `actual` untersucht. Wir trainieren stattdessen einen Random Forest (Leo Breiman: 2001) und wählen die 50 Features mit der größten *Feature Importance*.

```
In [71]: ▶ y_train = df_train.pop("actual")
          from sklearn.ensemble import RandomForestClassifier
          clf = RandomForestClassifier(max_depth=3)
          clf.fit(df_train, y_train);
```

```
In [55]: ▶ # 100% Accuracy auf den Trainingsdaten
          clf.score(df_train, y_train)

          # Hausaufgabe: Testdaten suchen!
```

Out[55]: 1.0

In der (Gini-) *Feature-Importance* wird für jedes Feature notiert, in welchem Baum es wie oft zum Splitten verwendet wurde. Es wird bei jedem Split notiert, wie sehr sich die Gini-Impurity verbessert. Diese Verbesserung wird über alle Splits gemittelt.

```
In [73]: ▶ # viele, aber nicht alle Features werden mit Feature Importance 0 bewertet
          clf.feature_importances_
```

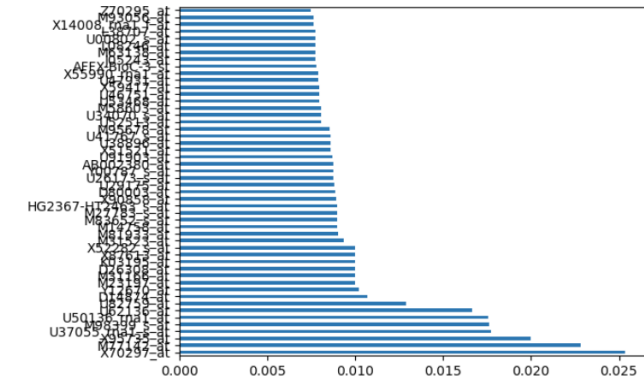
Out[73]: array([0., 0., 0., ..., 0., 0., 0.])

Nur ca. 200 Features haben eine von Null verschiedene Feature Importance:

```
In [74]: ▶ feat_imp = pd.Series ( clf.feature_importances_,
                                index=df_train.columns )
          feat_imp = feat_imp[feat_imp>0].sort_values()[::-1]
          feat_imp
```

```
Out[74]: X70297_at      0.025367
          M77142_at      0.022847
          X95735_at      0.020000
          U37055_rna1_s_at  0.017703
          M98399_s_at      0.017618
          ...
          U29953_rna1_at      0.001056
          D86972_at      0.001056
          HG2743-HT3926_s_at  0.001049
          D86963_at      0.001024
          M22632_at      0.001000
          Length: 195, dtype: float64
```

```
In [75]: ▶ # Visualisierung der "50 wichtigsten" Gene
          feat_imp[:50].plot.barh();
```



```
In [112]: ▶ # Die 50 wichtigsten Gene (nach Wichtigkeit absteigend sortiert)
          fi_names = feat_imp[:50].index
          fi_names
```

```
Out[112]: Index(['X70297_at', 'M77142_at', 'X95735_at', 'U37055_rna1_s_at',
                  'M98399_s_at', 'U50136_rna1_at', 'U62136_at', 'U82759_at', 'D14874_at',
                  'Y12670_at', 'M23197_at', 'M31166_at', 'D26308_at', 'K03195_at',
                  'X87613_at', 'X52282_s_at', 'M31523_at', 'M81933_at', 'M14758_at',
                  'M83652_s_at', 'M27783_s_at', 'HG2367-HT2463_s_at', 'X90858_at',
                  'D80003_at', 'U29175_at', 'U26173_s_at', 'Y00787_s_at', 'AB002380_at',
                  'U91903_at', 'X51521_at', 'U38896_at', 'U41767_s_at', 'M95678_at',
                  'U52513_at', 'U34070_s_at', 'M58603_at', 'U53468_at', 'U46751_at',
                  'X59417_at', 'U47931_at', 'X55990_rna1_at', 'AFFX-BioC-3_st',
                  'J05243_at', 'M63138_at', 'L08246_at', 'U00802_s_at', 'L38707_at',
                  'X14008_rna1_f_at', 'M93056_at', 'Z70295_at'],
                  dtype='object')
```

3.3 Hierarchisches Clustering der Genexpressionsdaten

Daten-Preprocessing wie im Paper: Daten werden *standardisiert*, so dass jede Spalte (=Genexpressionsdaten für ein Gen, indiziert mit 38 Patienten) Mittelwert 0 und Varianz 1 hat.

```
In [119]: df_train_std = (df_train - df_train.mean()) / df_train.std()
```

```
In [85]: df_train_std.mean(axis=0)
```

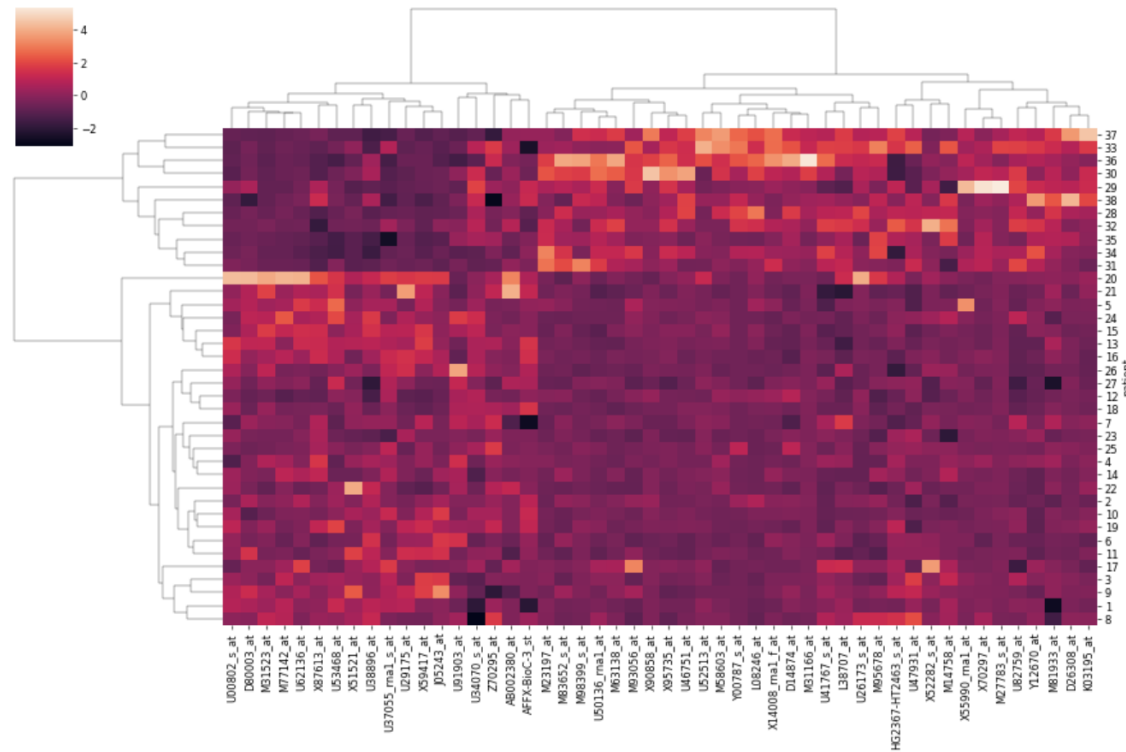
```
Out[85]: AFFX-BioB-5_at    -5.258951e-17  
AFFX-BioB-M_at    -1.694551e-16  
AFFX-BioB-3_at     1.899066e-17  
AFFX-BioC-5_at     5.258951e-17  
AFFX-BioC-3_at    -2.454177e-16  
...  
X83863_at         -4.674623e-17  
Z17240_at        -1.081007e-16  
L49218_f_at       0.000000e+00  
M71243_f_at       7.304099e-18  
Z78285_f_at      -5.843279e-18  
Length: 7129, dtype: float64
```

```
In [86]: df_train_std.std(axis=0)
```

```
Out[86]: AFFX-BioB-5_at     1.0  
AFFX-BioB-M_at     1.0  
AFFX-BioB-3_at     1.0  
AFFX-BioC-5_at     1.0  
AFFX-BioC-3_at     1.0  
...  
X83863_at          1.0  
Z17240_at          1.0  
L49218_f_at        1.0  
M71243_f_at        1.0  
Z78285_f_at        1.0  
Length: 7129, dtype: float64
```

Wende eine sog. Clustermap an ("doppeltes" hierarchisches Clustering auf Genen und auf Patienten):

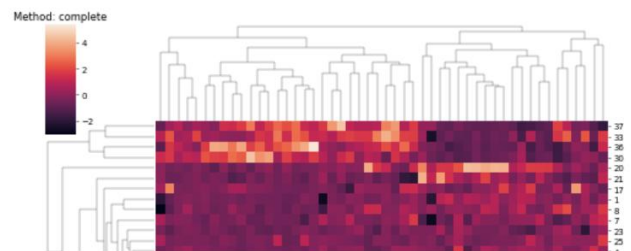
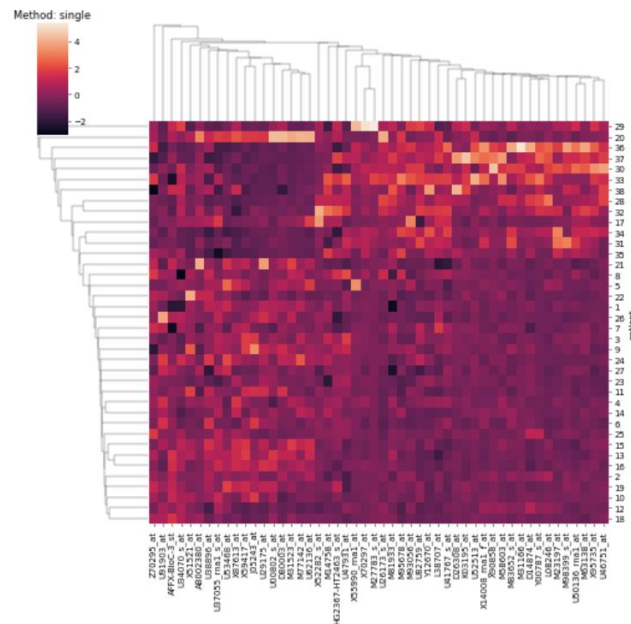
```
In [106]: ▶ sns.clustermap(df_train_std.loc[:,fi_names], method="ward", xticklabels=1, figsize=(15,10));  
plt.gcf().set_dpi(60)
```



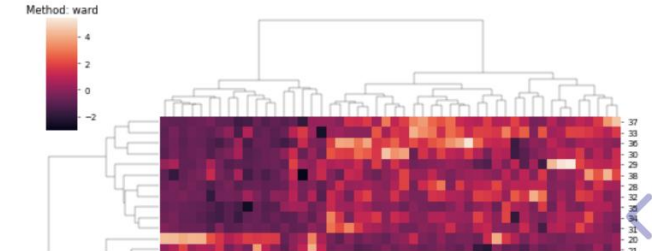
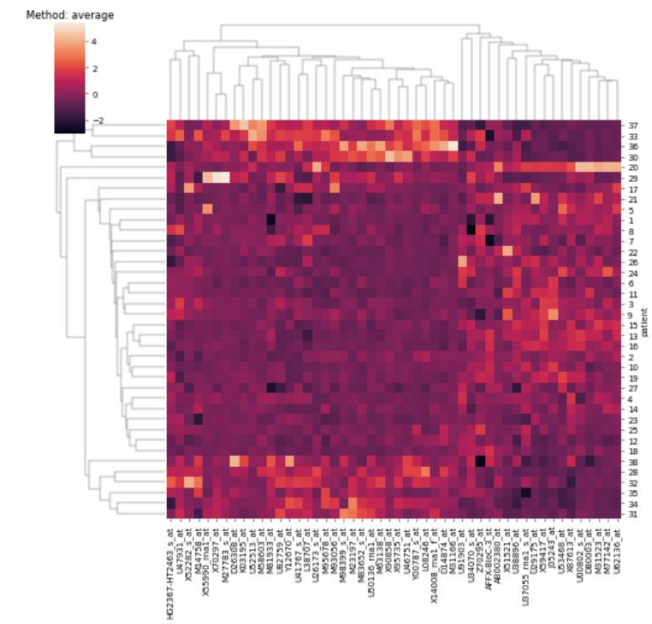
Die 11 AML-Patienten (#28-#38) werden im Personen-Dendrogramm (links) "auf oberster Ebene" einer anderen Gruppe zugeordnet als die ALL-Patienten. Ebenso sind zwei verschiedene Gruppen von Genen ersichtlich ("überexprimiert" vs. "unterexprimiert"), die im Gen-Dendrogramm (oben) "auf oberster Ebene" getrennt werden.

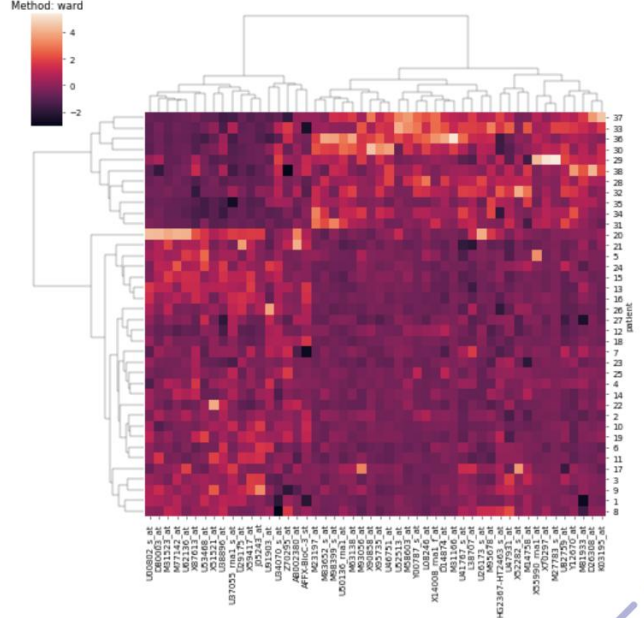
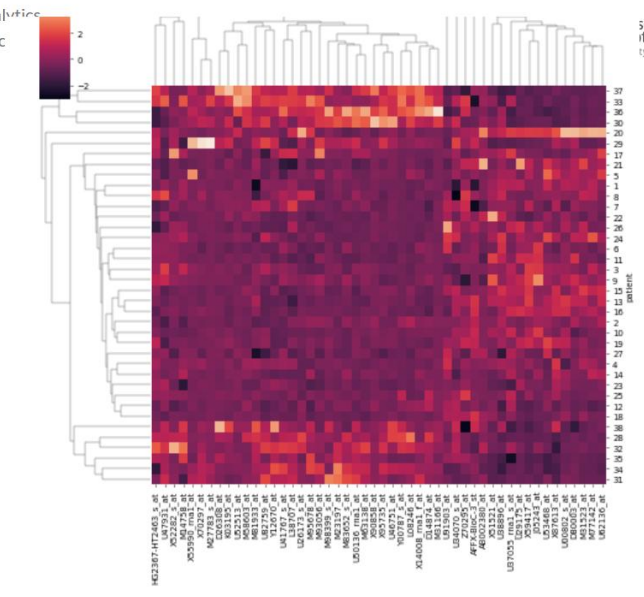
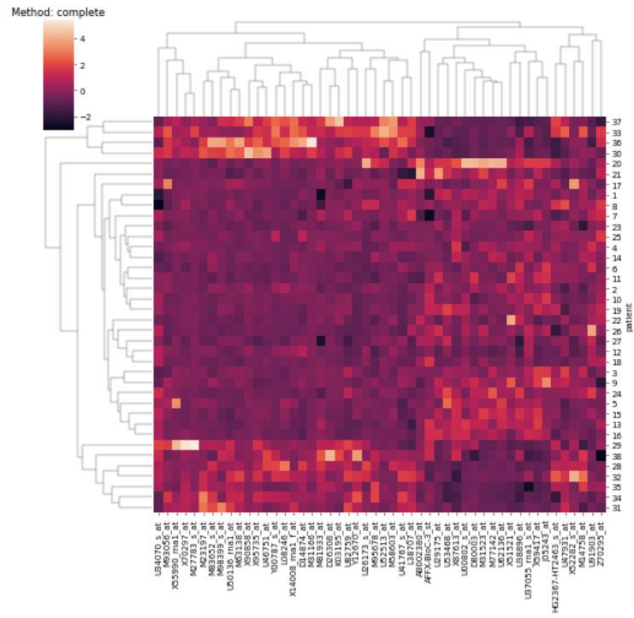
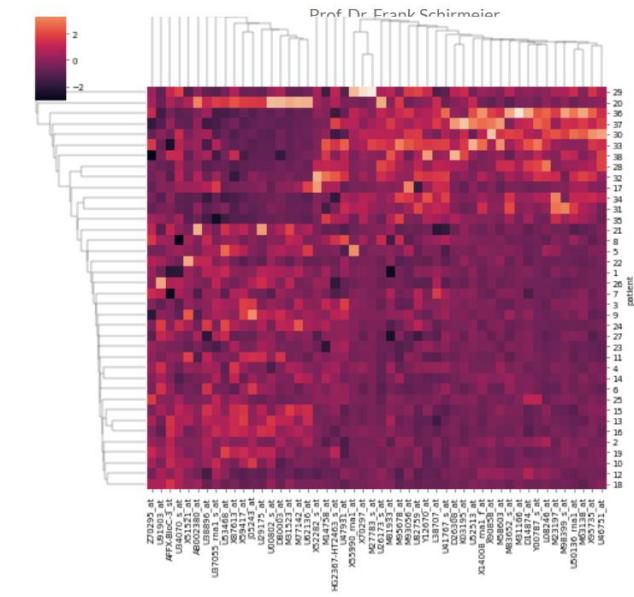
Nur bei `method="ward"` werden die 11 AML-Patienten (#28-#38) als einer Gruppe zugehörig erkannt:

```
In [115]: for method in ["single", "complete"]:  
sns.clustermap(df_train_std.loc[:,fi_names],  
method=method, xticklabels=1)  
plt.title("Method: "+method)  
plt.gcf().set_dpi(50)
```

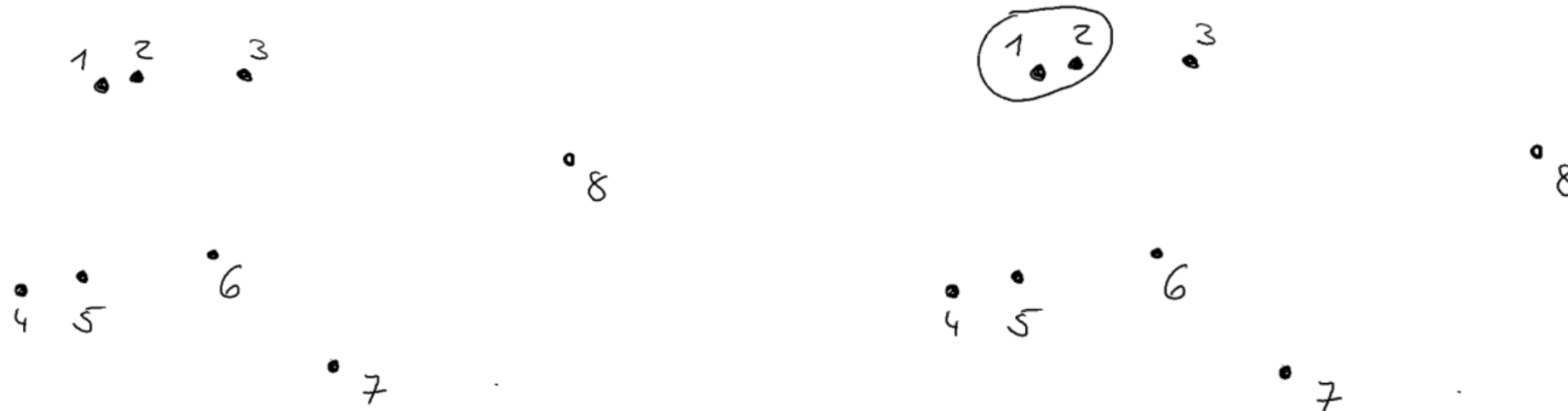


```
In [114]: for method in ["average", "ward"]:  
sns.clustermap(df_train_std.loc[:,fi_names],  
method=method, xticklabels=1)  
plt.title("Method: "+method)  
plt.gcf().set_dpi(50)
```





3.4 Hierarchisches Clustering (agglomerativ)



Man beginnt mit den Datenvektoren (hier 2d), zwischen denen ein gewisser **Abstandsbegriff** ("**Punktmetrik**") definiert sein muss.

Standardmäßig euklidischer Abstand:

$$d_{\text{euclid}}(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots}$$

Weise nun den beiden Punkten, die in der Punktmetrik den geringsten Abstand haben, einem gemeinsamen Cluster zu.

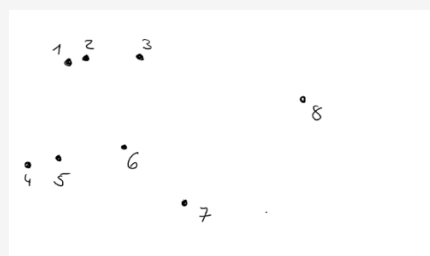
Um zu iterieren, muss der Abstand zu Clustern gemessen werden mittels einer "**Clustermetrik**".

Für die *Linkage*-Methode `method="single"` wird in der Clustermetrik die geringste Punktmetric zwischen Punkten aus den beiden Clustern verwendet:

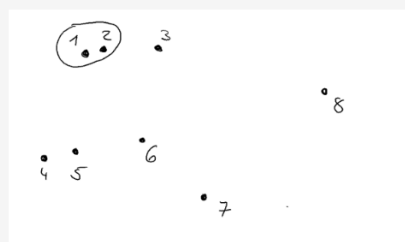
$$d_{\text{single}}(C_1, C_2) = \min_{x_1 \in C_1} \min_{x_2 \in C_2} d_{\text{Punkt}}(x_1, x_2)$$

Einzelne Punkte werden als ein-elementiges Cluster behandelt: $C = \{x\}$.

Schritt 1:



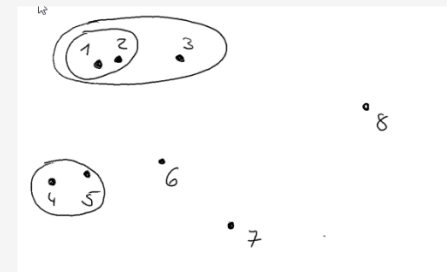
Schritt 2:



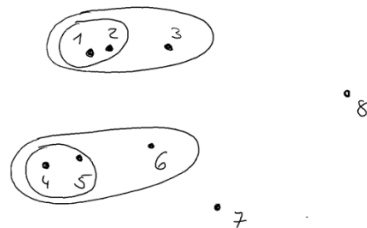
Schritt 3:



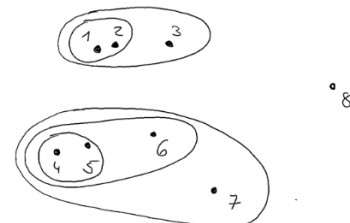
Schritt 4:



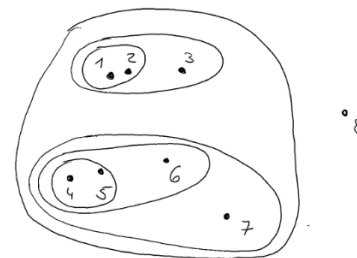
Schritt 5:



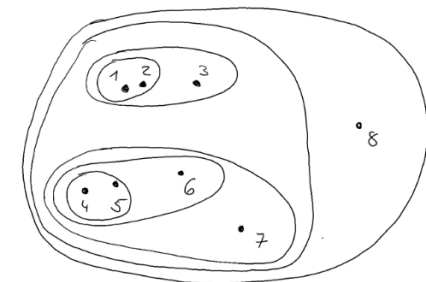
Schritt 6:



Schritt 7:

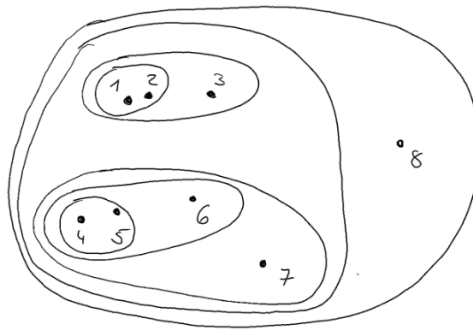


Schritt 8:

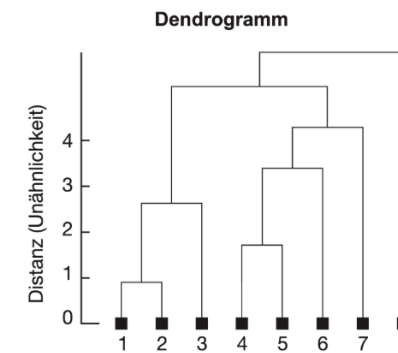


Das Ergebnis des Clustering-Verfahrens kann in einem Dendrogramm visualisiert werden:

Skizze:



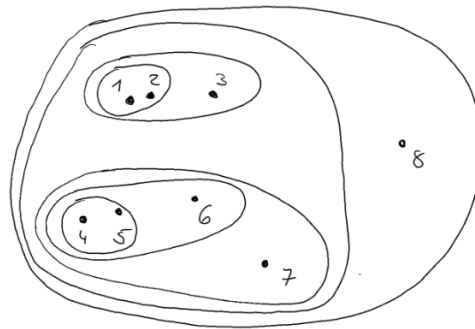
Dendrogramm:



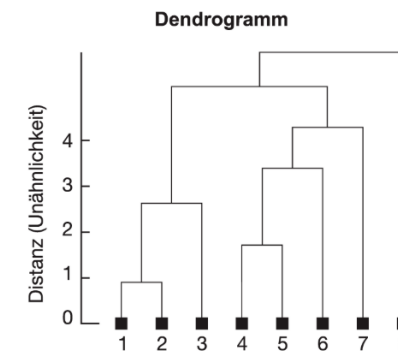
<https://wirtschaftslexikon.gabler.de/definition/dendrogramm-31008>

Das Ergebnis des Clustering-Verfahrens kann in einem Dendrogramm visualisiert werden:

Skizze:



Dendrogramm:



<https://wirtschaftslexikon.gabler.de/definition/dendrogramm-31008>

Vorteil gegenüber anderen Clustering-Verfahren (z.B. k-means): Man erhält ein **Kontinuum an Clustern** und kann sich nachträglich entschieden, mit wie viel Clustern man arbeiten möchte.

Mit den nachfolgenden Linkage-Methoden wird aus der Punktmetrik eine Clustermetrik abgeleitet:

- `method="single"` : Die kleinste Punktmetrik zwischen Punkten aus den beiden Clustern:

$$d_{\text{single}}(C_1, C_2) = \min_{x_1 \in C_1} \min_{x_2 \in C_2} d_{\text{Punkt}}(x_1, x_2)$$

- `method="complete"` oder `"maximum"` : Die größte Punktmetrik:

$$d_{\text{complete}}(C_1, C_2) = \max_{x_1 \in C_1} \max_{x_2 \in C_2} d_{\text{Punkt}}(x_1, x_2)$$

- `method="average"` : Der Durchschnitt der Punktmetriken zwischen Punkten aus den beiden Clustern:

$$d_{\text{average}}(C_1, C_2) = \frac{1}{C_1 \cdot C_2} \sum_{x_1 \in C_1} \sum_{x_2 \in C_2} d_{\text{Punkt}}(x_1, x_2)$$

In der Linkage-Methode `method="ward"` wird keine Clustermetrik betrachtet:

- `method="ward"` : Man beginnt mit einelementigen Clustern. Für jeden potentiellen nächsten Schritt ($C' \rightarrow C' \cup C''$) wird folgende Abweichung von den Clustermittelpunkten ("Varianz") berechnet:

$$V = \sum_C \sum_{x \in C} d_{\text{euclid}}(x, \bar{x}_C)^2, \text{ wobei } (\bar{x}_C)_i = \frac{1}{|C|} \sum_{x \in C} x_i$$

Der nächste Schritt mit der niedrigsten "Varianz" wird dann ausgeführt.

3.5 Beispiel: MNIST-Datensatz

```
In [122]: ▶ from sklearn.datasets import load_digits
           digits = load_digits( n_class=6 )
           target = digits["target"]
           data = digits["data"]
           target.shape, data.shape
```

Out[122]: ((1083,), (1083, 64))

```
In [147]: ▶ # Um das Ergebnis interpretieren zu können,
           # führe eine Dimensionsreduktion durch
           from sklearn.manifold import TSNE
           data_trf = TSNE().fit_transform(data)
           data_trf.shape
```

Out[147]: (1083, 2)

In `model.labels_` werden die gefundenen Cluster-IDs gespeichert:

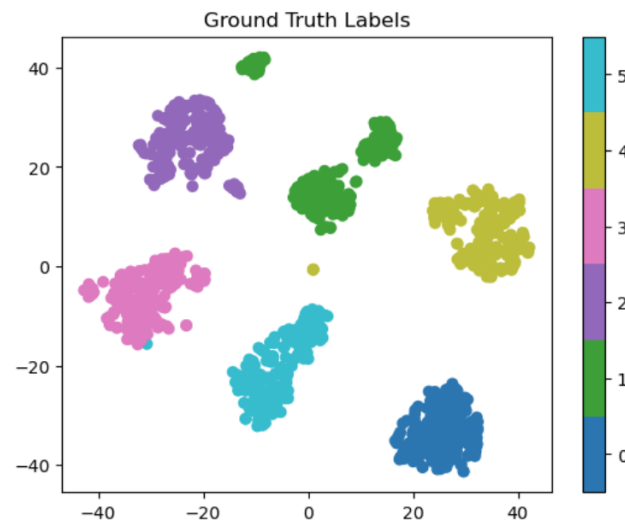
```
In [149]: ▶ from sklearn.cluster import AgglomerativeClustering
           model = AgglomerativeClustering(n_clusters=6)
           model.fit(data)

           model.labels_
```

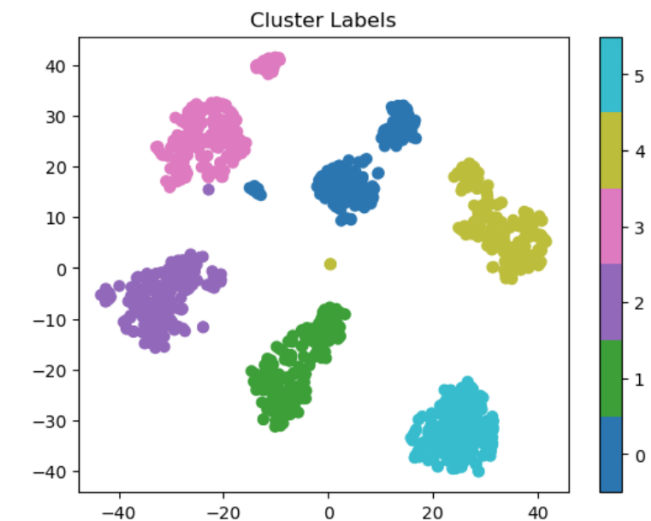
Out[149]: array([5, 0, 0, ..., 4, 4, 5], dtype=int64)

Visualisierung der dimensionsreduzierten Daten:

```
In [165]: ▶ plt.scatter (data_trf[:,0], data_trf[:,1],  
                        c=target,  
                        cmap=plt.get_cmap("tab10",6),  
                        vmin=-0.5, vmax=5.5 )  
cbar = plt.colorbar(ticks=np.arange(0,7,1) )  
plt.title("Ground Truth Labels")  
plt.show()
```



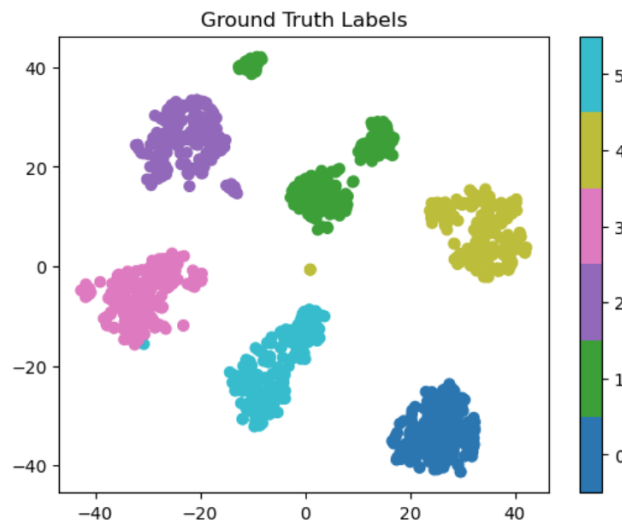
```
In [145]: ▶ plt.scatter (data_trf[:,0], data_trf[:,1],  
                        c=model.labels_,  
                        cmap=plt.get_cmap("tab10",6),  
                        vmin=-0.5, vmax=5.5 )  
cbar = plt.colorbar(ticks=np.arange(0,7,1) )  
plt.title("Cluster Labels")  
plt.show()
```



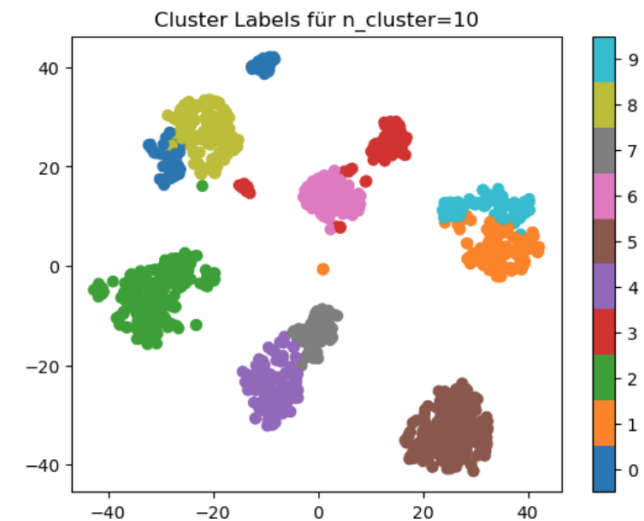
3.6 Wahl der "richtigen" Cluster-Zahl (n_cluster ist normalerweise nicht vorher bekannt)

```
In [152]: ▶ from sklearn.cluster import AgglomerativeClustering  
model = AgglomerativeClustering(n_clusters=10)  
model.fit(data);
```

```
In [153]: ▶ plt.scatter (data_trf[:,0], data_trf[:,1],  
                        c=target,  
                        cmap=plt.get_cmap("tab10",6),  
                        vmin=-0.5, vmax=5.5 )  
cbar = plt.colorbar(ticks=np.arange(0,7,1) )  
plt.title("Ground Truth Labels")  
plt.show()
```



```
In [158]: ▶ plt.scatter (data_trf[:,0], data_trf[:,1],  
                        c=model.labels_,  
                        cmap=plt.get_cmap("tab10",10),  
                        vmin=-0.5, vmax=9.5 )  
cbar = plt.colorbar(ticks=np.arange(0,10,1) )  
plt.title("Cluster Labels für n_cluster=10")  
plt.show()
```



Zeichne das Dendrogramm (für eine kleinere Datenstichprobe):

```
In [159]: ▶ # https://scikit-learn.org/stable/auto_examples/cluster/pl
def plot_dendrogram(model, **kwargs):
    from scipy.cluster.hierarchy import dendrogram

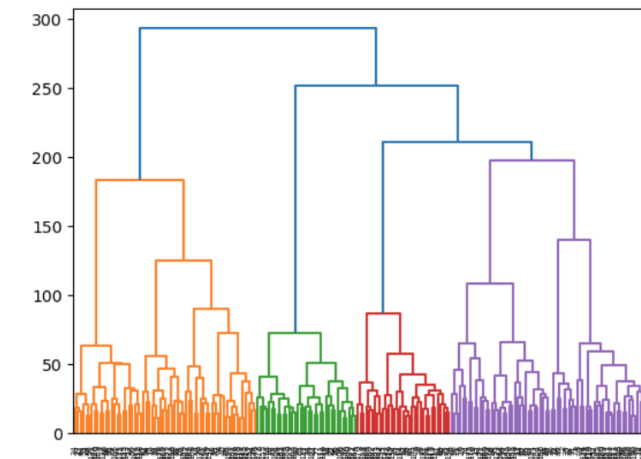
    # Create Linkage matrix and then plot the dendrogram

    # create the counts of samples under each node
    counts = np.zeros(model.children_.shape[0])
    n_samples = len(model.labels_)
    for i, merge in enumerate(model.children_):
        current_count = 0
        for child_idx in merge:
            if child_idx < n_samples:
                current_count += 1 # Leaf node
            else:
                current_count += counts[child_idx - n_samples]
        counts[i] = current_count

    linkage_matrix = np.column_stack(
        [model.children_, model.distances_, counts]
    ).astype(float)

    # Plot the corresponding dendrogram
    dendrogram(linkage_matrix, **kwargs)
```

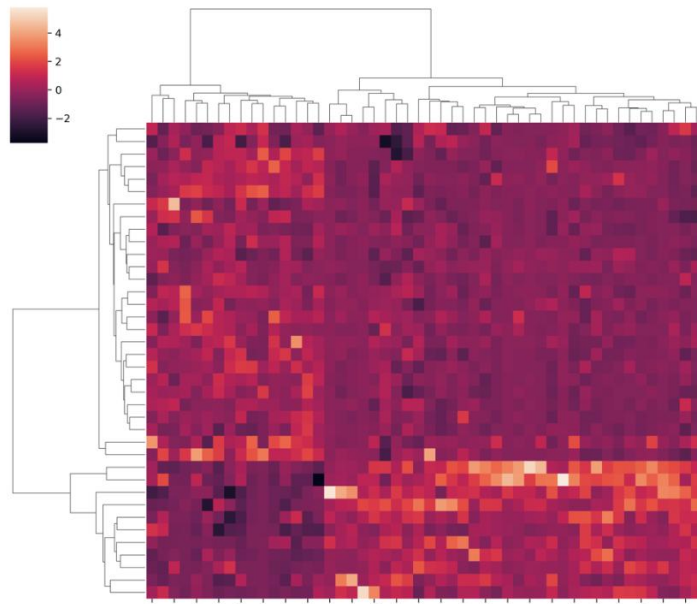
```
In [164]: ▶ from sklearn.cluster import AgglomerativeClustering
model = AgglomerativeClustering(
    distance_threshold=0, n_clusters=None)
model = model.fit(data[:200])
plot_dendrogram(model)
```



Anhand des Dendrograms kann versucht werden, einen "guten" Wert für `n_clusters` zu finden:

Auf welcher Höhe möchte man das Dendrogramm abschneiden (= "das Verschmelzen stoppen")?

Bei Höhe 150 würde man 6 Cluster erhalten. Diese Wahl ist aber **nicht eindeutig**.



1. Wiederholung

2. Personalisierte Medizin

3. Clusterverfahren

Vielen Dank für Ihre Aufmerksamkeit!